Towards a unified understanding of synaptic plasticity: parsimonious modeling and simulation of the glutamatergic synapse life-cycle

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Summary

Understanding how learning and memory formation work in the brain is a major challenge in neuroscience, with important implications for many other fields, including medicine and industry. It is nowadays widely accepted that *synaptic plasticity* is the biological foundation of these higher order brain functions. So far, many different plastic behaviors have been intensively studied and characterized, leading to the definition of several forms of plasticity: structural, functional, homeostatic, inhibitory, and many others. Unfortunately, despite all the interest and efforts of the scientific community, a complete and consistent understanding of synaptic plasticity is still lacking.

The main goal of this study is to unify data and theories on synaptic plasticity in a comprehensive model, suitable for studying learning and memory down to the synapse level. To reach our objective, we identified a minimal set of biological mechanisms responsible for plastic dynamics and integrated them into a single synapse model, relying whenever possible on well accepted sub-models from literature. We designed a data-driven fitting and generalization strategy to parameterize all excitatory-to-excitatory synapses in a large scale reconstruction of neocortical tissue [Markram et al., 2015]. Finally, we tested the effects of functional synaptic plasticity on neural circuits in simulations.

Our model was able to capture not only the outcome of Spike Timing Dependent Plasticity (STDP) protocols used in the training phase, but also the one of all others available in the same experimental dataset [Markram et al., 1997b]. Moreover, it correctly reproduced results on distance-dependent synaptic plasticity [Sjöström and Häusser, 2006], even though this dataset was never used during fitting. In network simulations, we observed the emergence of a self-regulatory homeostatic mechanism, preventing runaway excitation. Furthermore, we noticed the strengthening of connections between neurons that are similarly innervated, as previously shown in vitro [Perin et al., 2011].

Keywords: synaptic plasticity, rewiring, Long Term Potentiation (LTP), Long Term Depression (LTD), neural networks, learning, memory, neocortex, simulation, neuroscience, Blue Brain Project (BBP).

Sommario

Capire come l'apprendimento e la formazione della memoria lavorino nel cervello è uno dei principali obiettivi della neuroscienza, con importanti implicazioni per molti altri settori, tra cui la medicina e l'industria. È oggi ampiamente accettato che la *plasticità sinaptica* sia la base biologica di queste alte funzioni cerebrali. Finora sono stati studiati e caratterizzati diversi comportamenti plastici, portando alla definizione di svariate forme di plasticità: strutturale, funzionale, omeostatica, inibitoria e molti altri. Purtroppo, nonostante il grande interesse e gli sforzi della comunità scientifica, non esiste ancora una visione completa e coerente della plasticità sinaptica.

L'obiettivo principale di questo studio è quello di unificare i dati e le teorie sulla plasticità sinaptica in un modello completo, idoneo allo studio dell'apprendimento e della memoria fino al livello delle singole sinapsi. Per raggiungere il nostro obiettivo, abbiamo individuato un insieme minimo di meccanismi biologici responsabili delle dinamiche plastiche e li abbiamo integrati in un unico modello di sinapsi, affidandoci, ove possibile, a risorse ben note in letteratura. Abbiamo progettato una strategia di adattamento e generalizzazione per parametrizzare tutte le sinapsi eccitatorio-eccitatorie in una ricostruzione di tessuto neocorticale su larga scala [Markram et al., 2015]. Infine abbiamo testato gli effetti della plasticità sinaptica funzionale sui circuiti neurali in simulazione.

Il nostro modello è stato in grado di catturare non solo il risultato dei protocolli Spike Timing Dependent Plasticity (STDP) utilizzati nella fase di parametrizzazione, ma anche di tutti gli altri disponibili nello stesso set di dati sperimentali [Markram et al., 1997b]. Inoltre, ha riprodotto correttamente i risultati sulla plasticità sinaptica dipendente dalla distanza [Sjöström and Häusser, 2006], nonostante questo set di dati non fosse mai stato utilizzato durante la parametrizzazione. Nelle simulazioni di rete, abbiamo osservato l'emergere di un meccanismo di autoregolazione omeostatico, utile ad impedire il sovraeccitamento. Inoltre, abbiamo notato il rafforzamento delle connessioni tra neuroni innervati in modo simile, come precedentemente mostrato in vitro [Perin et al., 2011].

Parole chiave: plasticità sinaptica, rewiring, Long Term Potentiation (LTP), Long Term Depression (LTD), reti neurali, apprendimento, memoria, neocorteccia, simulazione, neuroscienze, Blue Brain Project (BBP).

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1 Introduction

"Learning never exhausts the mind."

Leonardo Da Vinci

Learning and memory are undoubtedly among the most critical and fascinating brain functions. Indeed complex organisms such as mammals would not survive in the external world without the capability to acquire new *knowledge* and to retrieve it whenever needed (i.e. to avoid predators). While this simple argument surely explains why learning and memory are *critical*, the *fascinating* aspect deserves deeper analysis. Let us consider the brain as a dynamical learning system, an engineering artifact. To begin with, our system is constantly bombarded by very heterogeneous stimuli from the external world: images, sounds, odors and so on. Of all the information received, very little matters. The important bits need to be identified, processed, stored and, most importantly, linked to existing knowledge, possibly of different nature (i.e. the sound of rain and the need of an umbrella). Furthermore, memory capacity is arguably limited and so it has to be managed in an effective way. That is, there must be space for new memories, but important ones have to be kept for the entire lifetime of the organism. Each step of this chain poses fundamental challenges to the design of such a system. Most of these challenges are nowadays unsolved and they represent active areas of research in engineering and science. If we now consider that not only the brain actually solves all of them, but also it does that in an efficient manner, we are left with one obvious question: *how?*

Understanding how learning and memory formation work in the brain would have important implications in many fields, including Medicine. Research on memory-related pathologies, such as the Alzheimer disease, could greatly benefit from a deeper understanding of information storage principles. Furthermore, the subject has also captured the interest of Industry, due to the recent successes of Artificial Neural Networks (ANN) and Deep Learning in many commercial applications, such as image and speech recognition [LeCun et al., 2015]. Although ANN are only loosely inspired to biological neural networks, certain features of biological learning represent the holy grail of modern Artificial Intelligence (AI). In particular, no AI-based system has ever been able to show strong generalization capabilities. That is, artificial systems have limited ability to adapt to unforeseen situations, as they are usually trained to perform a given set of tasks and any variation to the set requires re-training. Brain research is not the only way to look into these issues, but might represent the missing viewpoint to allow the design of next generation AIs.

The first step towards a better understanding of how learning and memory work is obviously to identify what are their biological building blocks. It is nowadays widely accepted that *synaptic plasticity*, the main subject of this work, is the biological substrate

of these higher order brain functions. A (chemical) synapse is a contact point between two neurons allowing the exchange of information[1]. Synapses are dynamic structures: they can form and dissolve, but also adapt their responses to neural activity and other environmental conditions. This set of capabilities is collectively referred to as *synaptic plasticity*. As synapses are one of the most important factors determining the response of neurons to stimuli and their behavior can change over time, they represent the ideal foundation for learning and memory.

So far many different plastic behaviors have been intensively studied and characterized, leading to the definition of several "forms" of plasticity, such as:

- **Functional plasticity**, the capability of neurons to modify the strength of existing connections. This form of adaptation have been the subject of many modeling studies in recent years.
- **Structural plasticity**, identified by a group of major changes of the brain microcircuit, ranging from creation or elimination of synapses to axonal or dendritic remodeling.
- **Homeostatic plasticity**, referring to a set of slow mechanisms regulating and maintaining the operating regimes of the neurons, despite ongoing plastic changes and turnover of the molecular substrate.

Regardless of this taxonomy, devised for technical and historical reasons, synaptic plasticity is nowadays considered an integrated phenomenon. Already in 1999, it was shown that persistent strengthening of synaptic connection, namely Long Term Potentiation (LTP), and formation of new synapses are strongly correlated [Toni et al., 1999]. Years later, also weakening of synaptic connections, or Long Term Depression (LTD), was found to be accompanied by important structural changes, synapse elimination in this case [Bastrikova et al., 2008]. The continuous accumulation of such evidences challenges the idea of a sharp distinction between forms of plasticity. The plethora of plastic changes observed in the brain may be the product of a single biological machinery, operating over different time scales.

Unfortunately, despite all the interest and efforts of the scientific community, a complete and consistent understanding of synaptic plasticity is lacking. In particular, there is still no consensus on what the most relevant plastic mechanisms are and how they are orchestrated together to actually enable learning.

The main difficulty to establish such a *unifying theory of synaptic plasticity* is the sparsity of available data. Lab experiments involving synaptic plasticity are inherently very complex. First of all, individual synapses are difficult to observe and even more difficult to control. For this reason, establishing any cause-effect link between network dynamics and synaptic variables is hardly feasible. Second, plastic mechanisms span

[1] In the classic view of synaptic transmission, information flows in one direction only, from the pre-synaptic neuron (the sender) to the post-synaptic one (the receiver). However, it is now known that synapses are capable of retrograde signaling and indeed these mechanisms are crucial for synaptic plasticity.

multiple time scales, from a millisecond to several days. Such a large range makes it extremely hard to understand how plastic mechanisms influence each other. Furthermore, experimental design is particularity difficult for the slowest time scales. This is indeed the main reason why plastic changes happening on different time scales were traditionally described as separate forms of plasticity. Last but not least, the heterogeneity of synaptic pathways suggests that the number of experiments required to quantitatively map the whole spectrum of plastic changes is currently beyond our reach.

The modeling approach has been proposed as a viable solution to overcome all the experimental limitations. Unfortunately, the lack of data has also limited our capability to build global theories on synaptic plasticity with a solid empirical foundation. As a consequence, most available models focus on very specific aspects only, such as LTP or homeostatic plasticity.

The main goal of this study was to unify data and theories on synaptic plasticity in a comprehensive model, suitable for studying learning and memory in cortical networks, down to the neuronal and synaptic level. We decided to restrict our scope to connections between excitatory neurons in the somatosensory cortex of the rat brain, as this system offers the largest datasets. Furthermore, connections between excitatory neurons accounts for 80% of synapses and almost %100 percent of spiny ones, which are thought to encapsulate the rich molecular machinery for plasticity. Particular attention has been payed to the identification of a minimal set of key plastic mechanisms and to the characterization of their interactions. The simulation framework, developed to complement this work, allows the observation of learning dynamics at the synapse level (i.e. LTP/LTD or synaptogenesis), neuron level (i.e. N-methyl-D-aspartate receptor (NMDAR) spikes or synapse cooperativity) and network level (i.e. emergence of connectivity motifs and structure-function relationships). For this reason, we propose this resource as a new tool for investigating learning and memory dynamics in the brain. Our hope is that this work will trigger more studies attempting to integrate data and models in the field of plasticity research. Moreover, we would value as a major achievement of this thesis the adoption of our simulation framework by other scientists to help the design of targeted in vitro and in vivo experiments.

To reach our objective, we identified a minimal set of biological mechanisms responsible for plastic dynamics and integrated them into a single synapse model, building from well accepted sub-models in literature. We then designed a two-step procedure for parameterizing the synapse model for any synaptic pathway. In the first step we use a custom Genetic Algorithm (GA) for fitting the parameters of experimentally characterized pathways to data. The second step generalizes the results of the previous, producing a predicted parametrization for uncharacterized pathways. The procedure was finally used to parameterize the plastic behavior of all 4 billion excitatory synapses in a large-scale computer reconstruction of neocortical tissue [Markram et al., 2015]. The reconstructed microcircuit was than simulated on the DOE ANL MIRA and BBPIV IBM BlueGene/Q supercomputers. To allow the parametrization and simulation of such a large piece of digital brain tissue, several technological challenges have been faced. The methods herein developed to solve them have, in principle, broad applicability in the fields of optimization, High Performance Computing (HPC) and simulation-based

science.

Over the course of this study, several artifacts have been produced. In particular, we designed and implemented:

i High-level model of the synapse lifecycle.

ii NEURON synapse model, and the corresponding parameterization, suitable to investigate synaptic plasticity in large scale simulations

iii Method to fit and generalize synaptic parameters from sparse data

iv Framework for synaptic plasticity modeling and simulation

The rest of this dissertation follows a bottom-up approach: we build from the experimental evidence on plastic mechanisms to the implications of synaptic plasticity on the structure and function of digitally reconstructed neural circuits. In Chapter 2 we summarize the key experimental results on synaptic plasticity and propose a parsimonious model of the synapse life-cycle. The goal of this chapter is to provide a short review of some key experimental results on synaptic dynamics, later used for model design. Chapter 3 describes how our synaptic model was developed, parametrized on available data and generalized to cover uncharacterized synaptic pathways. Chapter 4 presents the results of our first large-scale simulations of plastic neocortical networks. Here we describe the main effects of functional synaptic plasticity on neural circuits. We investigated the behavior of the network under two conditions: spontaneous activity and thalamocortical stimulation. Finally in chapter 5 we draw our conclusions, identifying the limitations of the present study and proposing future directions of research on the subject.

2 Physiology, anatomy and life cycle of synaptic connections

"In the case of all things which have several parts and in which the totality is not, as it were, a mere heap, but the whole is something beside the parts, there is a cause."

ARISTOTLE, *Metaphysics*

The main objective of this thesis was to unify data and theories on synaptic plasticity in a comprehensive synapse model. A systematic understanding of the different synaptic states and transformations was then a fundamental prerequisite for this work. Unfortunately, such a complete overview has not been proposed yet. To move the first steps in this direction, we derived a summarizing description of what we can call the *synapse life cycle* from the vast body of literature on synaptic plasticity.

In this chapter we present our view of the synapse life cycle and set the foundations for the synapse model proposed in chapter 3. We start by briefly introducing the main "components" of excitatory synapses and how they are orchestrated together to implement the synaptic functions (Sec. 2.1). We then focus on the characterization of the different synaptic states and their relationships, as emerging from our review of experimental literature (Sec. 2.2).

2.1 Structure and function of excitatory synapses

Synapses are biological electro-chemical transducers, required for information transmission and decoding in the brain. That is, they not only allow the propagation of electric signals from a neuron to another (transmission), but also affect the information content of the signal (decoding), as we will discuss later in this section.

Excitatory synapses consist of two parts, a presynaptic terminal (or bouton) and a postsynaptic specialization, bound together by a dense network of extracellular proteins in the so called synaptic cleft. The basic functioning of an excitatory synapse can be roughly summarized as follows. When an action potential (AP) is fired by the presynaptic neuron, the bouton might release one or more vesicles, containing neurotransmitter. Neurotransmitter molecules diffuse then in the synaptic cleft and activate the appropriate receptors, sitting on the postsynaptic side of the synapse. The activation of postsynaptic receptors causes an inward current which depolarizes the cell membrane, effectively propagating the electric signal form the presynaptic neuron to the postsynaptic one.

The magnitude and time course of the postsynaptic depolarization depend on several pre- and post-synaptic factors, such as: AP timing, location of the synapse on the dendritic arborization, extracellular ionic conditions, number of vesicles released and receptors available. For this reason, a synapse is not just propagating an electric pulse from a neuron to another, but it is actually "translating" a relatively poor signal, the AP, into a much richer representation, the postsynaptic depolarization. Considering that many of the factors responsible of signal decoding (i.e. the number of released vesicles) are actively adjusted by plastic mechanisms, we can reformulate the definition of synaptic plasticity as *the capability to dynamically adjust the readout of pre- and postsynaptic state*.

Let us now focus on the anatomy of a synapse, starting from the presynaptic side. As mentioned earlier, the bouton contains one or more vesicles, loaded with neurotransmitter. The most prominent neurotransmitter of excitatory cells is glutamate [1]. Vesicles ready to be released (or "docked") accumulate at the synaptic surface of the bouton, called the Active Zone (AZ). The machinery responsible for vesicle release is quite complex and still not completely understood, but in first approximation it consists of a set of Voltage Dependent Calcium Channels (VDCC), coupled with SNAP (Soluble NSF Attachment Protein) REceptor (SNARE) proteins. When an AP invades the bouton, the calcium influx through the VDCC activate the SNARE complex, causing vesicle fusion with the cell membrane and the consequent release of neurotransmitter. Vesicle release is a stochastic process. An AP may or may not trigger a release event, depending on several conditions, such as the availability of docked vesicles. The number of vesicles in the AZ and their probability of release are subject to short- and long-term forms of synaptic plasticity.

The vast majority of excitatory synapses are located on dendritic spines. Spines are small membrane protrusions, usually shorter than 2 µm and having a volume of approximatively 1 fL. They come in different sizes and shapes, ranging from thin cylindric formations to "mushroom" like structures, with a clear *neck* and *head*. With respect to synaptic plasticity, spines are extremely important for two reasons. First, they allow specificity of plastic changes through biochemical compartmentalization of signaling molecules, such as calcium ions. That is, several ion species cannot easily escape the spine, limiting the signal they convey to the synapses *in situ*. Second, they allow to bridge the distance between a presynaptic axon and a postsynaptic dendrite, tremendously extending the number of possible synaptic partners for a given neuron.

Most of the postsynaptic receptors are located on the spine head, right in front of the presynaptic AZ, packed in the so called Postsynaptic Density (PSD). The two most important receptors at excitatory synapses are alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate receptors (AMPARs) and N-methyl-D-aspartate receptors (NMDARs). AMPARs are the largest population of receptors and contribute most of the

[1] Strictly speaking no neurotransmitter is excitatory nor inhibitory. The effect of neurotransmitter release on the postsynaptic neuron depends on the type of receptors binding to it. However, considering that a cell can only produce a single type of neurotransmitter and that the number of different neurotransmitters and binding receptors in the brain is limited, the terms *excitatory cell* and *inhibitory cell* are quite common.

synaptic current under normal conditions. They are permeable to sodium and potassium ions, but not calcium. Insertion and removal of AMPARs is one of the most powerful, and noticeable, plastic change happening at excitatory synapses. NMDAR are undoubtedly the single most important protein for synaptic plasticity. At resting membrane potential, the NMDAR is blocked by magnesium ions. That is, even if bound to glutamate, the receptor does not carry any current. However, at depolarized potentials, the block is removed allowing the flux of several ion species, calcium in particular. Considering that postsynaptic depolarization can be due to both a backpropagating action potential (bAP) or a local electrical event such as the activation of a cluster of nearby synapses, the NMDAR is considered a *coincidence detector* of pre- and post-synaptic activity. In other words, two conditions are required to obtain an NMDAR-mediated current: the release of glutamate from the presynaptic terminal and the depolarization of the postsynaptic membrane. Furthermore, the order of these two events is non-commutative. Due to the latencies of the chemical and electrical dynamics of synaptic transmission, the release of neurotransmitter shortly followed by postsynaptic depolarization produces a much larger current than the reverse order of events. This property is extremely interesting from the computational point of view because, thanks to the NMDAR, a synapse can distinguish a causal relationship between the activity of the pre- and post-synaptic neuron form an anti-causal one.

Receptors are not the only channels inserted in the spine membrane. Voltage dependent ion channels are a major class of proteins related to several aspects of neurons physiology, from the electrical behavior to synaptic plasticity [Hille et al., 2001]. VDCCs are a large family of ion channels, responsible for a large fraction of the total calcium current in spines. Depending on their kinetic, VDCC can be classified as High Voltage Activated (HVA) and Low Voltage Activated (LVA) channels. HVA channels are particularly relevant for several forms of synaptic plasticity, given their activation as a consequence of large depolarizing events, such as a bAP.

In this brief overview of the anatomy and physiology of synapses we neglected several important components, such as metabotropic receptors or internal calcium stores. These elements are surely important for synapse functioning, but their role with respect to synaptic plasticity is either poorly understood or thought to be of secondary relevance. As mentioned earlier in this section, our goal is to distill the most important aspects of synaptic structure and function to produce a model suitable to study plasticity in large scale simulations.

2.2 Synapse life cycle

Based on our analysis of experimental literature, we distilled a simplified description of the synapse life cycle, shown in figure 2.1. We start by considering the process of synapse formation. This subject was extensively reviewed in two seminal publications [Holtmaat and Svoboda, 2009, Ziv and Garner, 2004] that guided our initial steps and provided the foundation for our analysis. We can identify two main paths for synaptogenesis: *ex novo* synapse formation and Multi Synapse Bouton (MSB) formation [Holtmaat and Svoboda,

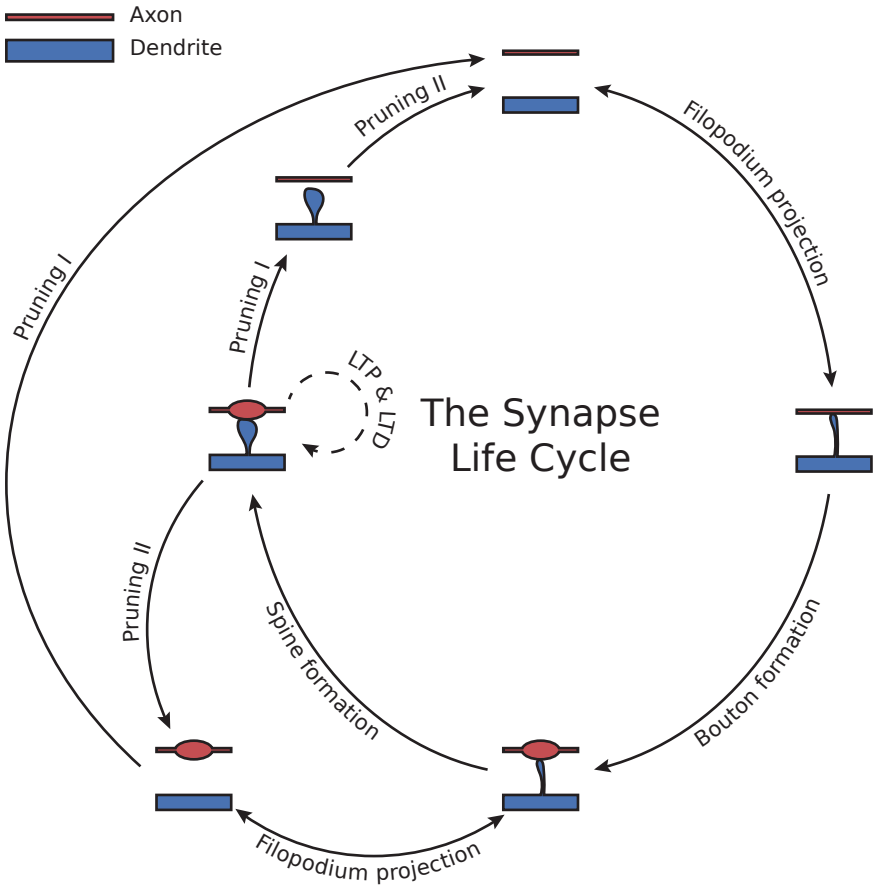


Figure 2.1 – Simplified model of the synapse life cycle. We consider six possible axo-dendritic configuration. Each solid arrow represent a plastic transition from one state to another. The dashed arrow identify functional changes at mature synapses, without major structural adaptation.

2009]. Let us consider the former case (Fig. 2.2). The initial requirement for any de novo synapse is a close axo-dendritic apposition that can be bridged by a spine [Fares and Stepanyants, 2009]. Spines are preceded by the elongation of filopodia, thin actin protrusions originating from the dendrites [Ziv and Smith, 1996, Kasai et al., 2010]. Filopodia have an extremely high turnover rate and often they get in contact with nearby axons before being eliminated [Ziv and Garner, 2004, Ziv and Smith, 1996]. Under particular circumstances [2], these axo-dendritic contacts can lead to the ordered maturation of the presynaptic (first) and postsynaptic (later) structures [Ziv and Garner, 2004, Friedman et al., 2000], and eventually to the formation of the so called "silent synapses". A (postsynaptically) silent synapse, is a synapse lacking a PSD despite a well formed AZ

[2] The mechanisms behind the stabilization of axo-dendritic contacts are mostly unknown, however, an interesting possibility is the involvement of mobile vesicles exocytosis [Ziv and Garner, 2004].

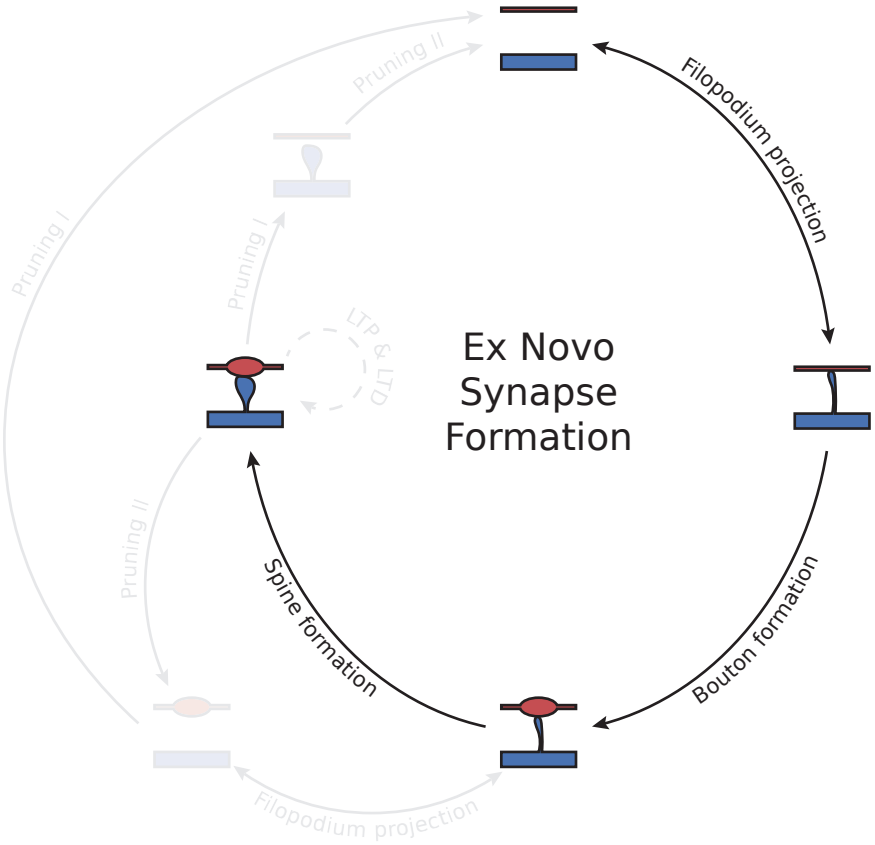


Figure 2.2 – Ex novo synapse formation. Filopodia can contact axons several microns away from their source dendrite. Occasionally, these events can lead to the formation of a presynaptic bouton and postsynaptic spine, followed by synapse maturation.

and, for this reason, not able to induce an Excitatory Post-Synaptic Potential (EPSP) [Kullmann, 2011]. However, silent synapses usually possess postsynaptic NMDARs and this is the key of their activity-dependent maturation, or "unsilencing"[Kullmann, 2011]. The ordered and close activation of the pre and postsynaptic neurons releases the magnesium block from the NMDARs, allowing the flux of ions, including calcium [Kullmann, 2011]. The calcium influx causes the insertion of AMPARs in the postsynaptic membrane and so the formation of a PSD at the new born functional synapse [Kullmann, 2011].

Another possible pathway for synaptogenesis is what can be defined as "bouton recycling" (Fig. 2.3). It was shown that filopodia, and spines, tend to project towards existing boutons and form synapses [Holtmaat and Svoboda, 2009, Knott et al., 2006]. The steps of this model of synaptogenesis are similar to those described above, with

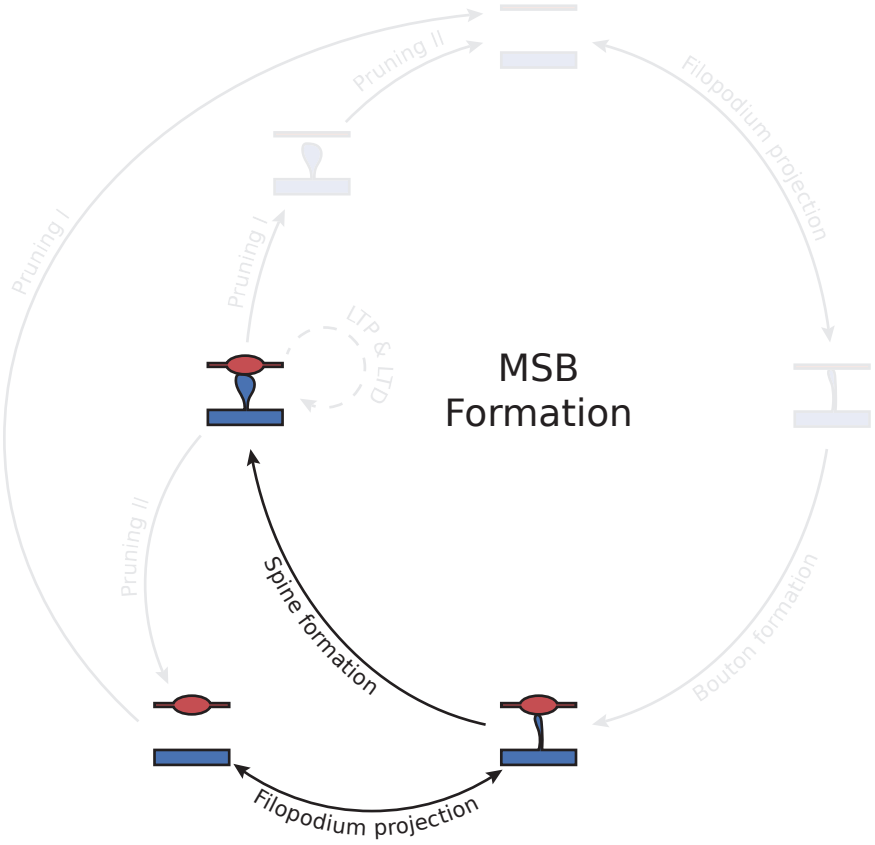


Figure 2.3 – Multi synapse bouton formation. An existing bouton can be contacted by a newly formed spine. Weather this configuration can be maintained indefinitely or not is currently unknown.

the main difference that the process ends with the creation of a MSB [Holtmaat and Svoboda, 2009]. Nevertheless, some interesting insights deserve to be mentioned. First, glutamate might be responsible of the attraction of filopodia and spines to existing boutons [Holtmaat and Svoboda, 2009, Richards et al., 2005], supporting the idea that the exocytosis of mobile vesicles might be crucial for the *de novo* synapse formation. Second, the coexistence of more than one synapse on the same bouton in not necessarily a stable condition: synapses on the same bouton might compete for presynaptic resources, causing the pruning of one or more of them [Holtmaat and Svoboda, 2009, Toni et al., 2007]. Last, MSBs potentially resolve the discrepancies found in the turnover rates of spines and boutons [Holtmaat and Svoboda, 2009, De Paola et al., 2006, Holtmaat et al., 2005].

After the analysis of synaptogenesis, we now focus on functional synapses (Fig. 2.4). Mature synapses show, among all the other synaptic states, the richest spectrum of

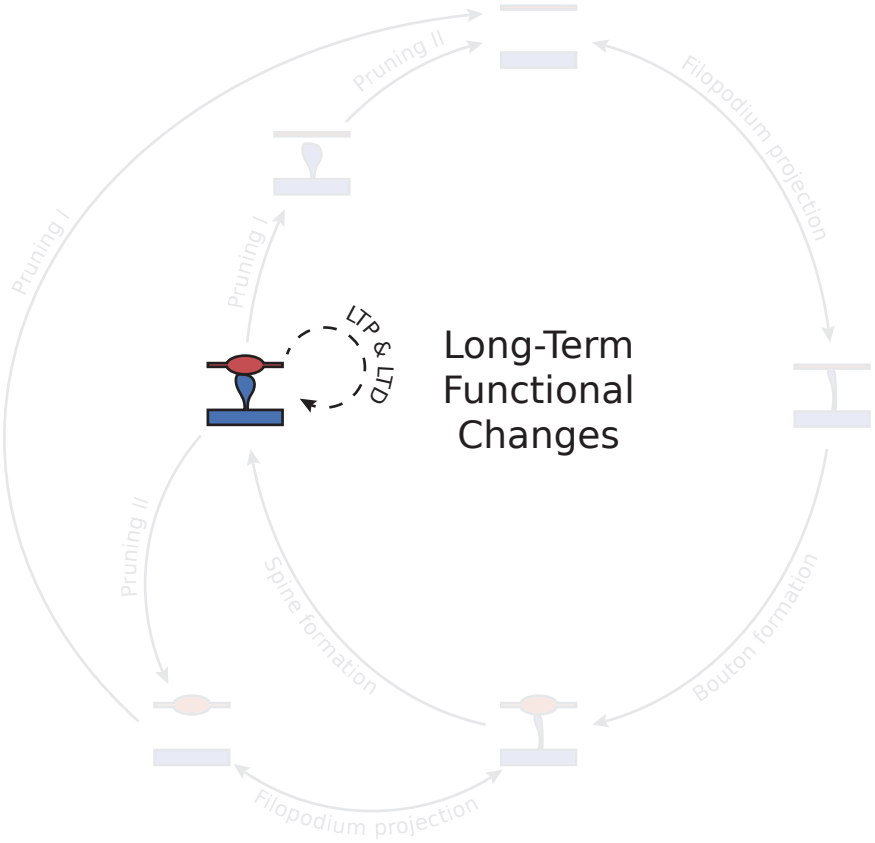


Figure 2.4 – Long-term functional changes. Mature synapses can transiently or persistently adapt their response to presynaptic activation. Persistent changes, such as LTP (LTD), do not require substantial structural adaptation, although they are often accompanied by the enlargement (shrinkage) of spine and bouton.

plastic dynamics, spanning from morphological to functional changes. Long Term Potentiation (LTP) and Long Term Depression (LTD) are among the most striking of these dynamics. The term LTP refers to a persistent increase of the postsynaptic response due to the activation of the presynaptic fibers [Bliss and Lømo, 1973]. Without going into the details of the expression and maintenance of LTP (for a review see [Malenka and Bear, 2004]), we briefly point out that LTP can have a presynaptic locus of expression [Markram and Tsodyks, 1996] or a postsynaptic one or both [Sjöström et al., 2007]. Postsynaptic forms of LTP usually involve the insertion of AMPARs in the postsynaptic membrane (but see [Bliss and Collingridge, 1993]). Remarkably enough, LTP plays also an important structural role. It was indeed shown that LTP and synaptogenesis are strongly related [Holtmaat and Svoboda, 2009, Toni et al., 1999], raising the attention of

the scientific community about the connections between structural and functional plasticity. LTD is the specular process of LTP and can be oversimplified as a drop in release probability [Sjöström et al., 2007] or the removal of AMPAR from the membrane, with the consequent reduction of the synaptic response [Malenka and Bear, 2004]. One of the most interesting and curious things about LTD is its relationship with synapse elimination. Since at least one decade, LTD is speculated to be correlated with spine pruning (for a review see [Segal, 2005]) and, ultimately, to be one of the causes of synapse elimination. However, as remarked by M. Segal, the conjecture of a link between LTD and spine pruning is mainly due to theoretical reasoning rather than experimental evidence: the well established connection between spinogenesis and LTP, plus the complementarity of this last process with LTD, have suggested the logical conclusion that LTD and spine elimination must be correlated, but nobody has proved it yet [Segal, 2005].

Despite the disruptive power of plastic mechanisms, morphological variables at functional synapses appear to be tightly correlated. Harris and Stevens [1989] showed that PSD area is strongly correlated with several pre- and post-synaptic variables, such as spine head volume, bouton volume and number of vesicles. Furthermore, spine head volume is also correlated with the number of vesicles [Harris and Stevens, 1989] and strong correlation between PSD area and spine head volume has been reported across several cortical and non-cortical areas [Harris and Stevens, 1989, Schikorski and Stevens, 1999, Arellano et al., 2007]. Such findings suggest the presence of remarkable regulatory mechanisms at the level of single synapses. As many forms of long-term plasticity have well defined morphological correlates, we can speculate that eventually any punctual change will be matched over time by the whole synapse. For example, we could reasonably expect that even a purely postsynaptic form of LTP, expressed through the rapid insertion of AMPAR, will be followed by presynaptic adaptation (i.e. an increase in size of the Readily-Releasable Pool (RRP)).

Spine (bouton) pruning is the last structural change needed to produce a consistent model of the synapse life cycle (Fig. 2.5). In recent years, a large effort has been devoted to identify the link between LTD and synapse elimination. Several experiments have shown that there is indeed a correlation between spine retraction and LTD [Nägerl et al., 2004, Segal, 2005]. Although spine retraction does not necessary imply synapse elimination [Segal, 2005], the discovery of a connection with LTD prompts us to wonder about the relationships between functional and structural changes at depressed synapses. In particular, the exact order of the structural changes leading to synapse elimination is still a matter of debate, being that a consequence of LTD or not. While for many years the trend was to consider the retraction of spines the hallmark of synapse elimination, recent papers demonstrated a substantial involvement of presynaptic and extracellular structures, leading to the idea that synapse elimination might start with bouton loss and only later be followed by spine vanishing [Bastrikova et al., 2008, Becker et al., 2008]. For the moment we still cannot completely rule out any of these hypotheses and perhaps the two dynamics might coexist in the brain.

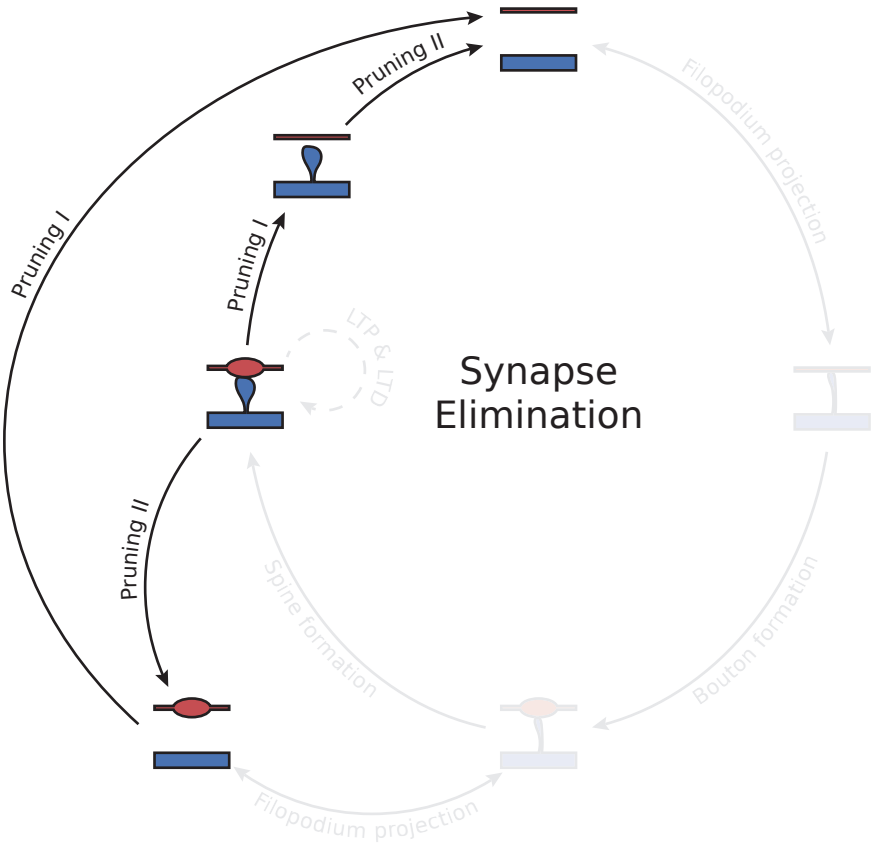


Figure 2.5 – Synapse elimination. Two possible paths for pruning: presynaptically initiated or postsynaptically initiated. In the former case, bouton loss (pruning I) is followed by spine retraction (pruning II). In the latter case, the order of events is reversed. The two processes are likely to be independent.

3 Modeling and parameterization of excitatory synapses

"The devil is in the details."

ANONYMOUS

The most tangible contribution of this work to the field of Neuroscience was the development of a novel synapse model, suitable for studying the effects and implications of synaptic plasticity in the context of biophysically detailed simulations of neocortical tissue [Markram et al., 2015]. In the future, such a model could be used to investigate higher order brain functions, such as learning, in a multiscale manner from networks down to single neurons and synapses.

Synapse modeling is an active research topic bridging different fields, from biology to mathematics and computer science. In recent years many aspects of synaptic physiology have been successfully captured in a variety of theoretical frameworks. However, a single coherent description of the most relevant synaptic dynamics for learning and memory is yet not available. To fill this gap, we designed a novel synapse model to meet the following requirements:

i Accounting for all the main physiological features of excitatory synapses;

ii Constrained to experimental data whenever available;

iii Computationally parsimonious;

iv Coupled with a constrained network and a fitting infrastructure;

v Built reusing existent and consolidated results on synapse modeling whenever available;

vi Easily extensible as new data on neuronal and synaptic physiology and plasticity emerges.

The result is a simple but comprehensive implementation of the basic *synapse life cycle*, as presented in Chapter 2, for the NEURON simulator [Carnevale and Hines, 2006].

In this chapter we introduce the synapse model, which we will refer to as the *GluSynapse*, and describe all the steps and data sources required to parametrize it. In Section 3.1 we present all the sub-components of GluSynapse and how they are orchestrated together. Section 3.2 is dedicated to the fitting strategy specifically designed to parametrize the long-term plastic dynamics. Validation and predictions of the parametrized model

are then discussed in Section 3.3. Finally, in Section 3.4 we deal with some technical aspects of the NMODL implementation (NEURON) and simulation.

3.1 The GluSynapse model

GluSynapse is a collections of sub-models, as illustrated in Figure 3.1. Briefly, the model has two structural states: *functional* and *potential*. The former represents a regular excitatory synapse, while the latter describes a close apposition between an axon and a dendrite, suitable for hosting a synapse but not yet fully realized. Transitions from one structural state to the other are regulated by the rewiring mechanism (see Sec. 3.1.7).

In the functional state, presynaptic vesicle release (see Sec. 3.1.2) induces a postsynaptic current by opening the alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate receptors (AMPARs) and N-methyl-D-aspartate receptors (NMDARs) (see Sec. 3.1.1). Furthermore, Voltage Dependent Calcium Channelss (VDCCs) can also allow current fluxes if the cell membrane is sufficiently depolarized (see Sec. 3.1.3). We assumed that all excitatory synapses are located on spines (see Sec. 3.1.4), although our results can be easily generalized to shaft synapses. Postsynaptic calcium dynamics are explicitly taken into account (see Sec. 3.1.5), as they drive long-term changes of synaptic efficacy (see Sec. 3.1.6). Calcium ions enters the spine mostly through NMDAR and VDCC conductances at these synapses [Koester and Sakmann, 1998, Sabatini and Svoboda, 2000, Sabatini et al., 2002]. Release from internal stores and diffusion from dendrites was not taken into account in the model as their contribution to the total calcium current is still debated [Bloodgood and Sabatini, 2007]. Finally, synapse elimination is described as a stochastic process, dependent on synaptic efficacy (see Sec. 3.1.7).

In the potential state, the synapse model cannot induce any current into the parent dendrite. The only enabled component is the rewiring model to allow dynamic synapse formation (see Sec. 3.1.7).

The basic components of the functional state, excluding synaptic plasticity, were designed, parameterized and implemented by several scientists at the Blue Brain Project (BBP) over the last 12 years. The following sections provide a global overview of the whole model, emphasizing the original contributions of this work.

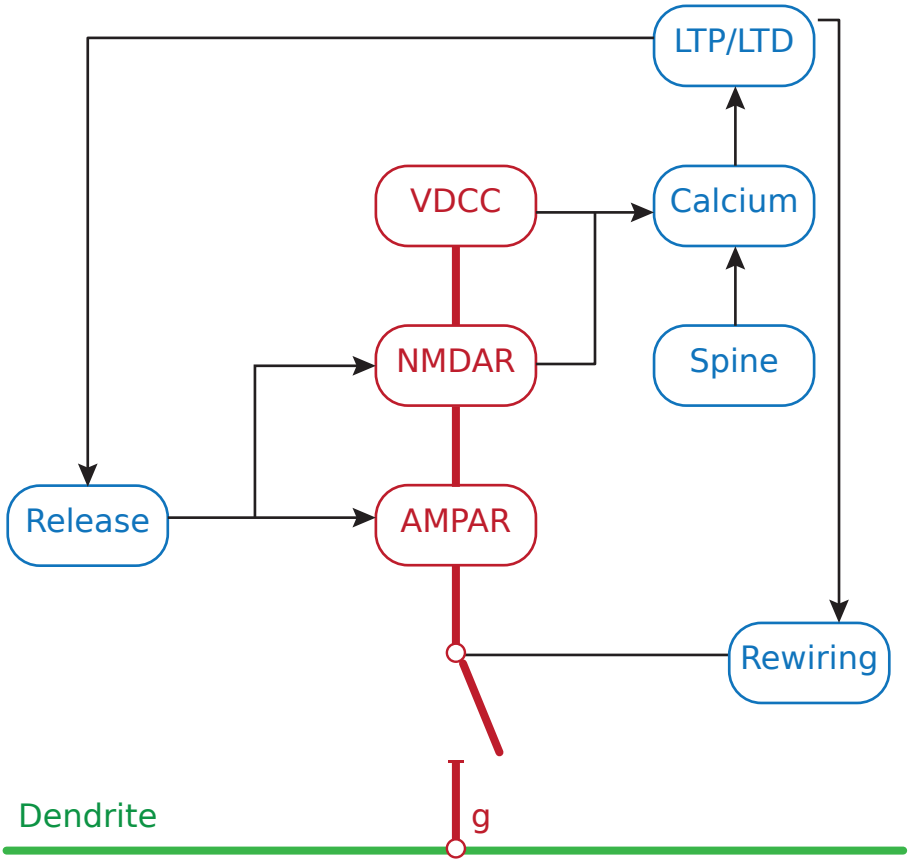


Figure 3.1 – GluSynapse logical structure. AMPAR and NMDAR with double exponential conductance profile and magnesium block dynamics (NMDAR only). Vesicle release mechanisms based on the Tsodyks-Markram (TM) model. VDCC of High Voltage Activated (HVA) family, based on R-type channel. Spine as an independent biochemical compartment. Calcium dynamics accounting for spine volume and simplified buffering/extrusion processes, bounded calcium dynamics as leaky-integration of free calcium. Long-term plasticity featuring both NMDAR-dependent Long Term Potentiation (LTP) and Long Term Depression (LTD). Rewiring mechanism supporting synaptogenesis and efficacy-dependent pruning.

3.1.1 AMPAR and NMDAR

AMPARs and NMDARs are among the most fundamental components of an excitatory synapse, as their activation depolarizes the postsynaptic neuron, and indeed they were already part of the BBP's model of neocortical tissue before the beginning of this work [Ramaswamy et al., 2012, Markram et al., 2015, Ramaswamy et al., 2015]. We did not alter the available models nor their parameterization in any way, but for the sake of

completeness we present them here.

The AMPAR and NMDAR conductances are commonly described using a double exponential profile:

$$g = g_{\max}(b - a) \tag{3.1}$$

$$\dot{a} = -\frac{1}{\tau_r}a + \text{jump} \cdot \delta(t - t_{\text{rel}}) \tag{3.2a}$$

$$\dot{b} = -\frac{1}{\tau_d}b + \text{jump} \cdot \delta(t - t_{\text{rel}}) \tag{3.2b}$$

$$\text{jump} = -e^{-\text{tp}/\tau_r} + e^{-\text{tp}/\tau_d} \tag{3.3a}$$

$$\text{tp} = \frac{\tau_r \tau_d}{(\tau_d - \tau_r) log(\tau_d/\tau_r)} \tag{3.3b}$$

where $g_{\max}$ (nS) is the peak conductance of the receptor population; a is the rising component of the conductance, with time constant τ_r (ms); b is the decaying component of the conductance, with time constant τ_d (ms); t_{rel} is the time of a successful release event (ms); tp (ms) is the time to peak of the conductance; jump is a normalization factor such that when $t = t_{\text{rel}} + \text{tp}$, $g = g_{\max}$.

The current generated by the synaptic population of AMPAR is then simply computed as

$$i_{\text{AMPAR}} = g \cdot (V_m - E) \tag{3.4}$$

while the one produced by the NMDARs depends also on the magnesium block dynamics

$$\text{mg} = \frac{1}{1 + e^{-c_1 V_m} c_2} \tag{3.5a}$$

$$i_{\text{NMDAR}} = \text{mg} \cdot g \cdot (V_m - E) \tag{3.5b}$$

where E (mV) is the reversal potential of the receptor; V_m (mV) is the membrane potential; c_1 and c_2 are appropriate constants (see Jahr and Stevens [1990]).

3.1.2 Vesicle release

Vesicle release is a complex electrochemical mechanism allowing information to flow from the presynaptic neuron to the postsynaptic one. Moreover, it is responsible for the distinctive features of synaptic transmission, such as *short-term facilitation* and *short-term depression*.

These dynamics are often described using the so called TM model [Tsodyks and Markram, 1997]. In the original formulation, the TM model consists of a set of three Ordinary Differential Equations (ODEs), designed and parameterized to describe the short-term dynamics of depressing synaptic connections, such as those between Thick-tufted pyramidal cells (TTPC) in layer 5. Over the course of the years, the model

has been refined and extended, for example to capture other short-term effects such as facilitation or to model independent release sites rather than whole connections (see [Fuhrmann et al., 2002, Loebel et al., 2009]). The latter formulation is particularly interesting because it reconciles the TM model with the quantal model of vesicle release [Purves et al., 2014].

As part of the BBP's effort to accurately reproduce synaptic transmission, Muller et al. [2011] further elaborated on the previous works to design a model of single release site dynamics featuring:

1. No resource consumption in case of transmission failure,
2. Explicit vesicle recovery dynamics,

while retaining convergence to canonical TM model for number of trials $\to \infty$. In this model, a release site R is assumed to be in one of two states: *ready* or *recovering*. Whenever R is in the *ready* state, the arrival of a presynaptic spike at time t triggers vesicle release with probability $U(t)$ given by:

$$U(t) = U(t_{\text{syn}}) \cdot e^{-(t-t_{\text{syn}})/\tau_{\text{fac}}} + U_{\text{SE}} \cdot (1 - U(t_{\text{syn}}) \cdot e^{-(t-t_{\text{syn}})/\tau_{\text{fac}}}) \tag{3.6}$$

where U_{SE} is the absolute release probability, that is, the release probability of a synapse not activated in a very long time; t_{syn} (ms) is the time of the last presynaptic spike; where τ_{fac} is the facilitation time constant. In case of successful release event, the postsynaptic AMPARs and NMDARs are activated as shown in Eq. 3.2 for $t_{\text{rel}} = t$) and the R state is set to *recovering*. Vesicle recovery is modeled as a survival process, where the probability of remaining in state *recovering* at time t P_{surv} is given by:

$$P_{\text{surv}}(t) = e^{-(t-t_{\text{syn}})/\tau_{\text{dep}}} \tag{3.7}$$

where τ_{dep} is the depression time constant.

We extended the model in Muller et al. [2011] by adding support for Multi Vesicular Release (MVR) [Rahmon et al., in preparation], a key feature of synaptic transmission with major implications on plasticity[1]. In our model, a pool of release sites is assumed to be available at every synapse. The dynamics of each release site are described as in Muller et al. [2011]. The activation functions of AMPAR and NMDAR, presented in Eq. 3.2), were adapted to ensure that the peak conductance is obtained only when all available vesicles are released:

$$\dot{a} = -\frac{1}{\tau_r} a + \text{jump} \cdot \frac{N_{\text{rel}}}{N_{\text{RRP}}} \cdot \delta(t - t_{\text{rel}}) \tag{3.8a}$$

$$\dot{b} = -\frac{1}{\tau_d} b + \text{jump} \cdot \frac{N_{\text{rel}}}{N_{\text{RRP}}} \cdot \delta(t - t_{\text{rel}}) \tag{3.8b}$$

[1] To give an example, LTP requires calcium influx through the NMDAR [Markram et al., 1997b] (see also Sec. 3.1.5 and Sec. 3.1.6). As vesicle release is required to activate theNMDAR, the number of vesicles ready to be released has a strong impact on the capability of individual synaptic contacts to potentiate.

where N_{rel} is the number of released vesicles at time t_{rel} and N_{RRP} is the size of the readily releasable pool.

As of the other sub-components of GluSynapse designed before the beginning of this work, parameters for the release model are available online through the Neocortical Microcircuit Collaboration (NMC) portal [Ramaswamy et al., 2015]. The sole exception is the number of readily releasable vesicles per synapse N_{RRP}, fixed to 2 from our preliminary results on MVR at excitatory connections [Rahmon et al., in preparation]. Although this value is not yet finalized at the time of writing, recent experimental findings on pyramidal to interneuron connections [Molnár et al., 2016] seem to support our choice.

3.1.3 VDCC

VDCCs are the most important source of calcium at dendritic spines after the NMDARs [Sabatini and Svoboda, 2000]. The model of VDCCs used in this work was designed and parametrized by Christian Rössert [Chindemi et al., in preparation], using data and components from several sources [Magee and Johnston, 1995, Sabatini and Svoboda, 2000]. In brief, a simple but inactivating R-type VDCC was designed in the Hodgkin-Huxley (HH) formalism. The population conductance is given by

$$g = g_{\max} m^2 h \tag{3.9}$$

$$\dot{m} = \frac{m_\infty - m}{\tau_m} \tag{3.10a}$$

$$\dot{h} = \frac{h_\infty - h}{\tau_h} \tag{3.10b}$$

$$m_\infty = \frac{1}{1 + e^{\frac{V_{hm} - V_m}{k_m}}} \tag{3.11a}$$

$$h_\infty = \frac{1}{1 + e^{\frac{V_{hh} - V_m}{k_h}}} \tag{3.11b}$$

where $g_{\max}$ (nS) is the peak conductance of the VDCCs population; m is the activation variable, with time constant τ_m (ms) and steady state m_∞; h is the inactivation variable, with time constant τ_h (ms) and steady state h_∞; V_{hm} (mV) is the half-maximum activation voltage and k_m (mV) is the slope factor; V_{hh} (mV) is the half-maximum inactivation voltage and k_h (mV) is the slope factor; V_m (mV) is the membrane potential, as usual.

The current induced by the population of VDCCs is then computed as

$$i = g(V_m - E_{\text{ca}}) \tag{3.12}$$

where E_{ca} is the reversal potential for calcium.

3.1.4 Spine

The vast majority of excitatory synapses are located on dendritic spines. With respect to synaptic plasticity, spines are particularly important because they act as biochemical compartments, granting specificity of synaptic changes. Furthermore, spine volume is a key determinant of intracellular calcium concentration, the main driver of synaptic plasticity.

In this work we do not explicitly model spines for computational reasons (see Sec. 3.4), but we do account for biochemical compartmentalization and volume-related effects on calcium concentration. Specifically, a spherical spine head is assumed for all computations and calcium ions are not allowed to diffuse into the parent dendrite.

Spine volume parameterization was carried out by Christian Rössert [Chindemi et al., in preparation], using data from Ruth Benavides-Piccione (unpublished at the time of writing). A crucial aspect of the parameterization was the correlation introduced between spine volume and NMDAR conductance [Arellano et al., 2007]. This feature produced normalized calcium transients across synapses. Consequently, we could easily generalize the long-term plasticity model parameters obtained for a prototypical synapse to all the others, as later explained in Sec. 3.2. The algorithm used to generate correlated pairs of spine volume and conductance is described in elsewhere [Chindemi et al., in preparation].

3.1.5 Postsynaptic calcium

The postsynaptic calcium model was designed and implemented by Christian Rössert using data from several sources [Sabatini et al., 2002, Cornelisse et al., 2007]. In brief, calcium current can flow through cell membrane via two paths: NMDARs and VDCCs. After entering the spine, calcium ions quickly bind to endogenous buffers and only a small fraction remains free. Slower mechanisms, such as calcium pumps and diffusion, are in place to re-establish the intracellular calcium concentration. These dynamics are all summarized in a single ODE, modeling free calcium at the postsynaptic membrane surface:

$$\frac{d[\mathrm{Ca}]_i}{dt} = (i^{\star}_{\mathrm{NMDAR}} + i_{\mathrm{VDCC}})\frac{\gamma}{V} - \frac{([\mathrm{Ca}]_i - [\mathrm{Ca}]_i^{(0)})}{\tau_{\mathrm{Ca}}} \tag{3.13}$$

where $i^{\star}_{\mathrm{NMDAR}}$ is the calcium component of the NMDAR-mediated current, as later described; i_{VDCC} is the VDCC-mediated current; γ is the fraction of free (non buffered) calcium; V is the spine volume; $[\mathrm{Ca}]_i^{(0)}$ is the intracellular calcium concentration at rest; τ_{Ca} is the time constant of calcium transients.

While VDCC current can all be considered calcium, NMDAR are permeable to several other ions, including sodium and potassium. The fractional calcium current P_f through the NMDAR can be calculated from the Goldman-Hodgkin-Katz (GHK) flux equation [Schneggenburger et al., 1993]:

$$P_f = \frac{i_{\mathrm{Ca}}}{i_{\mathrm{Ca}} + i_M} = \frac{4[\mathrm{Ca}]_o}{4[\mathrm{Ca}]_o + \frac{p_M}{p_{\mathrm{Ca}}}[M](1 - e^{2\frac{V_m F}{RT}})} \tag{3.14}$$

where i_{Ca} and i_M are respectively the current due to calcium ions and the one due to all monovalent ions; $[\mathrm{Ca}]_o$ is the extracellular calcium concentration; $[M]$ is the concentration of monovalent ions, assumed to be identical inside and outside the cell membrane; p_{Ca} and p_M are respectively the permeability to calcium ions and the one to monovalent ions; V_m is the membrane voltage; T is the temperature; F is the Faraday's constant and R is the ideal gas constant.

Unfortunately Eq. 3.14 cannot be directly used in our model as a scaling factor for the total NMDAR current because it loses physical meaning in the limit of $V_m \to 0$ mV, a threshold often crossed during action potential (AP) (see Fig. 3.2). That is, $P_f \to 1$ for $V_m \to 0$, but as also $i_{\mathrm{NMDAR}} \to 0$, the total calcium current would be incorrectly estimated. Based on the observation that the calcium current approaches 0 for $V_m \geq 40$ mV [Schneggenburger et al., 1993], we approximated the calcium component of the NMDAR conductance $g^{\star}_{\mathrm{NMDAR}}$ as a separate ion channel with reversal potential $E_{\mathrm{rev}} = 40$ mV:

$$g^{\star}_{\mathrm{NMDAR}} = \alpha P_f^{(-\infty)} g_{\mathrm{NMDAR}} \tag{3.15a}$$

$$i^{\star}_{\mathrm{NMDAR}} = g^{\star}_{\mathrm{NMDAR}}(V_m - E_{\mathrm{rev}}) \tag{3.15b}$$

where α is an appropriate scaling factor, determined by comparing model simulation results to Eq. 3.14 (see Fig. 3.2); $P_f^{(-\infty)}$ is the fractional calcium current at very negative potentials (Eq. 3.14, all other parameters as in Schneggenburger et al. [1993]); E_{rev} was assumed to be independent of extracellular calcium concentration, at least for values around 1 mM (see [Jahr and Stevens, 1993]). The proposed model was then tested against the findings of Schneggenburger et al. [1993] and accepted as a good approximation in the physiological voltage range and at relevant extracellular calcium contention (see Fig. 3.2).

Intracellular calcium dynamics are key for the induction of long-term synaptic changes and so they have been heavily studied in the context of Spike Timing Dependent Plasticity (STDP) [Nevian and Sakmann, 2006]. Even tough it is nowadays clear that calcium is the main driver for such changes, how exactly synapses could *read* this signal is still debated. One of the most (ab)used theories is the so called *calcium control hypothesis*: LTP is induced by large calcium transients, while LTD is tirggered by prolonged medium-level transients. Although this simple explanation is very elegant and appealing from the theoretical point of view, its experimental validation has proven to be particularity tricky and many open questions remain. In particular, it was shown that high-frequency stimulation triggers LTP while low-frequency does not [Markram et al., 1997b], but whether the former causes calcium accumulation at the synapse is unknown [Manninen et al., 2010]. Some experimental evidence seems to suggest that small-capacity and fast-binding calcium reaction partners need to be saturated before the mechanisms responsible of LTP and LTD could be activated (for a review see [Manninen et al., 2010]). Indeed preliminary analysis showed that calcium does not accumulate in our model at high-frequency stimulation (data not shown). For this reason, we speculate that the frequency dependence of LTP could not be explained based on free-calcium only and requires to consider the different calcium reaction partners. To account for such dy-

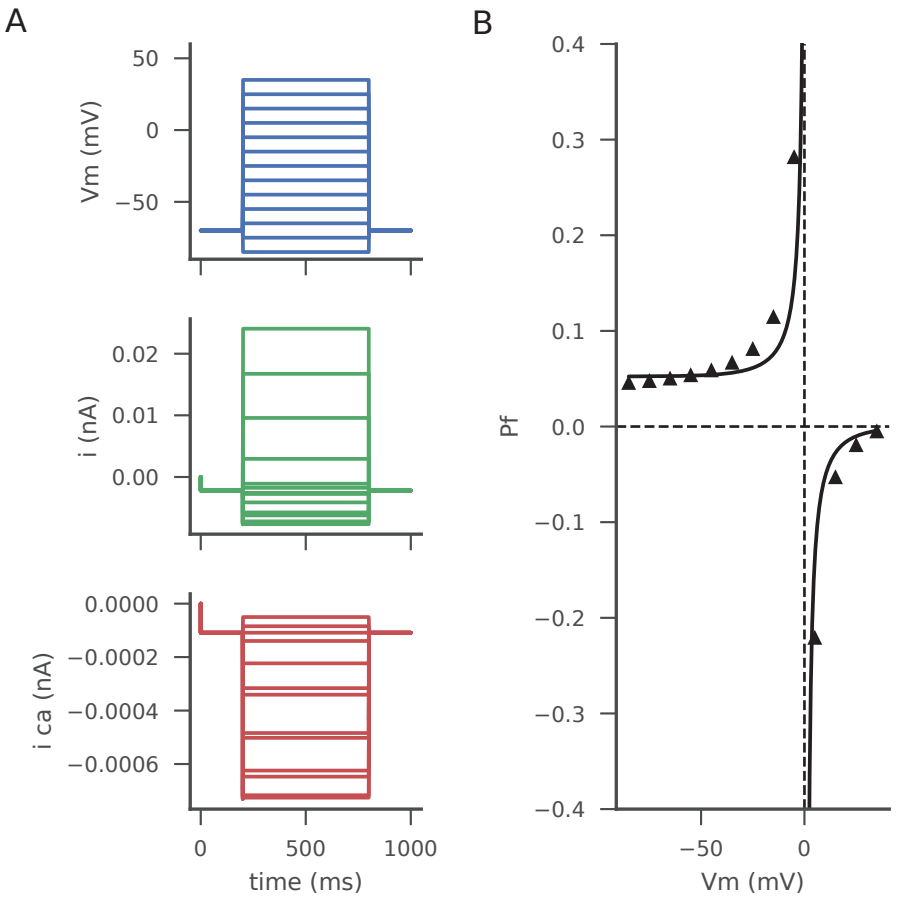


Figure 3.2 – Fractional NMDAR-mediated calcium current. A top, a series of voltage steps has been applied to a simulated patch of membrane around our model of NMDAR; the total NMDAR current (middle) and its calcium component (bottom) have been measured during voltage steps; persistent activation of the NMDAR was simulated by setting the decay time constant τ_d in Eq. 3.2 to an extremely large value. B, comparison of simulated fractional calcium current P_f (triangles) and predicted values from the GHK equation (solid line) [Schneggenburger et al., 1993]; simulated P_f was computed by dividing the value of calcium current by total current at 750 ms as reported in A.

namics without explicitly modeling the endogenous buffers, we introduced the calcium integrator $[\mathrm{Ca}]_i^\star$:

$$\frac{d[\mathrm{Ca}]_i^\star}{dt} = -\frac{[\mathrm{Ca}]_i^\star}{\tau_\star} + [\mathrm{Ca}]_i - [\mathrm{Ca}]_i^{(0)} \tag{3.16}$$

where $\tau_\star$ is an appropriate time constant for calcium integration, here fixed to 200 ms. This value was chosen considering both the average duration of an NMDAR-mediated spike [Schiller et al., 2000], considered the key signal for triggering LTP, and the typical stimulation window of LTP-inducing protocols [Markram et al., 1997b]. The idea behind

the introduction of $[\mathrm{Ca}]_i^\star$ is to prevent a single and isolated calcium transient from inducing any long-term change of synaptic efficacy, while allowing several pulses at high frequency, such as those used in STDP protocols, or large cooperative events, such as glsnmdar-mediated spikes, to produce enough activation to trigger a long-term synaptic change.

3.1.6 Long-term plasticity

As discussed in chapter 2, calcium-dependent forms of synaptic plasticity are the most studied and well characterized, both experimentally and theoretically. For this reason, we decided to adapt a calcium-based model of STDP by Graupner and Brunel [2012] to account for more realistic synaptic dynamics, such as those described above. Before discussing the details of the extended model, we briefly introduce the main features of the original one [Graupner and Brunel, 2012], from now on simply the Graupner-Brunel (GB) model.

The GB model assumes bistable synapses [Petersen et al., 1998, Bagal et al., 2005, O'Connor et al., 2005]. That is, the "efficacy" of each synapse is a dynamical variable with only two stable states: potentiated and depressed. Synaptic plasticity is then a transition from one state to the other [2]. State transitions are governed by postsynaptic calcium transients, through a thresholding mechanism. Whenever calcium crosses the depression (potentiation) threshold, the synaptic efficacy steps toward the depressed (potentiated) state. The magnitude of the steps depends on the time spent above the threshold.

The synaptic efficacy is an abstract concept of the model, to be mapped into an actual synaptic quantity (i.e. density of AMPARs, release probability or number of docked vesicles). In the original model, the efficacy is linearly translated to a generic "weight", a correlate of the peak synaptic conductance.

In equations, the Graupner and Brunel [2012] model is formulated as:

$$\begin{aligned}\tau\frac{d\rho}{dt} &= -\rho(1-\rho)(\rho_\star-\rho)+\gamma_p(1-\rho)\Theta[c(t)-\theta_p] \\ &\quad -\gamma_d\rho\Theta[c(t)-\theta_d]+\mathrm{Noise}(t)\end{aligned} \tag{3.17a}$$

$$\frac{dc}{dt} = -\frac{c}{\tau_{\mathrm{ca}}}+C_{\mathrm{pre}}\sum_i\delta(t-t_i-D)+C_{\mathrm{post}}\sum_j\delta(t-t_j) \tag{3.17b}$$

$$w = w_0+\rho(w_1-w_0) \tag{3.17c}$$

where ρ represents the synaptic "efficacy", with time constant τ; $\rho_\star$ delimits the basin of attraction of the two stable states; Θ is the Heaviside function; θ_d and θ_p are respectively the depression and potentiation thresholds; γ_d and γ_p the depression and potentiation rate; Noise(t) and activity dependent noise; $c(t)$ is the postsynaptic calcium concentration, with time constant tau_{ca}; C_{pre} is the calcium transient caused by a presynaptic

[2] Bistable synapses should not be confused with binary synapses (i.e. having only two possible values). In a bistable model, the synaptic efficacy is a continuous variable having two stable fixed points. That is, in the absence of synaptic activity the efficacy moves toward the closest fixed point.

spike at time t_i, the delay D account for the slow activation of the NMDAR; C_{post} is the calcium transient caused by a postsynaptic spike at time t_j; w is the synaptic weight; w_0 and w_1 are the weight of the depressed and potentiated state respectively. Another important variable of the GB model is the ratio of depressed synapses β, controlling the maximum potentiation (depression) obtainable:

$$\Delta\text{EPSP}_d = \frac{w_0}{\beta_0 w_0 + (1-\beta_0) w_1} \tag{3.18a}$$

$$\Delta\text{EPSP}_p = \frac{w_1}{\beta_0 w_0 + (1-\beta_0) w_1} \tag{3.18b}$$

where ΔEPSP_d and ΔEPSP_p are respectively the maximum amount of depression and the maximum amount of potentiation; β_0 is the initial ratio of depressed synapses; w_0 and w_1 have the usual meaning. For a more precise description of the GB model and its parameters, we invite the reader to refer to the original article by Graupner and Brunel [2012].

Our first modification of the GB model was dropping the noise term in the synaptic efficacy (Eq. 3.17a), as we already account for several stochastic aspect of synaptic transmission and membrane potential fluctuations. Second, we replaced the linear calcium component (Eq. 3.17b) with the more detailed description described in Sec 3.1.5, as NMDARs and VDCCs are explicitly modeled and so their contribution to postsynaptic calcium concentration. For more details on the postsynaptic calcium model please refer to Sec. 3.1.5. Finally, the abstract synaptic efficacy (Eq. 3.17a) is mapped into presynaptic release probability rather than a generic weight. We decided to restrict ourselves to presynaptic changes as they seem to be the predominant plastic variable in the somatosensory cortex [Tsodyks and Markram, 1997]. However, postsynaptic changes were indeed observed in the somatosensory cortex and other neocortical areas [Sjöström et al., 2007, Perin and Chindemi, in preparation] and they could be easily added to the model in the future. To account for the slow implementation of synaptic changes often observed in vitro, where the synaptic "strength" right after the induction protocol remains basically unaltered (see Fig. 3C in Markram et al. [1997b]), we introduced a delay factor in the efficacy to release probability mapping:

$$\frac{dU_{\text{SE}}}{dt} = \frac{U_{\text{SE}}^{(\text{target})} - U_{\text{SE}}}{\tau_{\text{change}}} \tag{3.19a}$$

$$U_{\text{SE}}^{(\text{target})} = U_{\text{SE}}^{(d)} + \rho[U_{\text{SE}}^{(p)} - U_{\text{SE}}^{(d)}] \tag{3.19b}$$

where U_{SE} is the absolute release probability, as previously discussed (see Sec. 3.1.2); τ_{change} is the time constant of the delay; $U_{\text{SE}}^{(\text{target})}$ is the target absolute release probability; $U_{\text{SE}}^{(d)}$ is the absolute release probability in the depressed state; $U_{\text{SE}}^{(p)}$ is the absolute release probability in the potentiated state.

As we mentioned before, we included a new term, the calcium integrator $[\text{Ca}]_i^\star$, to explain the frequency dependence of LTP. Now that we introduced the fundamentals of the GB model, we can clarify why this term was needed. It is known that pairing pre- and

post-synaptic activity at high frequency induces LTP, while low frequency is ineffective at triggering any change [Markram et al., 1997b]. In the GB model, potentiation and depression depend on the time spent by the postsynaptic calcium concentration about the respective thresholds θ_p and θ_d. However, as

1. layer 5 TTPC connections have strongly depressing short-term dynamics [Markram et al., 1997a],
2. we do not expect them to have a large Readily-Releasable Pool (RRP) of vesicles (see [Molnár et al., 2016]),
3. the $[\mathrm{Ca}]_i$ time constant τ_{Ca} is fast (see Sec. 3.1.5),

high-frequency stimulation could not produce significant calcium accumulation at these synapses and low-frequency stimulation is much more effective at triggering calcium events (Fig. 3.3 and 3.4). For this reason, comparing the time spent by the $[\mathrm{Ca}]_i$ above any threshold leads us to the conclusion that a different "read-out" would be needed to explain why high-frequency stimulation is more effective than low-frequency to induce LTP. The results of previous studies indicate that saturation of fast calcium buffers could be a requirement for LTP's frequency-dependence [Holmes and Levy, 1990, Markram et al., 1998]. The calcium integrator was the chosen read-out as it rewards a series of calcium events in a short time, somehow mimicking the interplay of a fast and a slow calcium buffer. Indeed using $[\mathrm{Ca}]_i^\star$ rather than $[\mathrm{Ca}]_i$ sorts times above threshold in the expected order (Fig. 3.4) and allowed us to fit the synapse model to LTP and LTD data (see Sec. 3.2).

Fitting our extended model of long-term plastic changes is a non trivial task. The main difficulty is the inaccessibility of the relevant synaptic variables, not directly observable during standard plasticity experiments, neither in vivo nor in vitro. For this reason, we designed of a special procedure to fit single synapses on the basis of whole connection data from STDP experiments in vitro, described in Sec. 3.2.

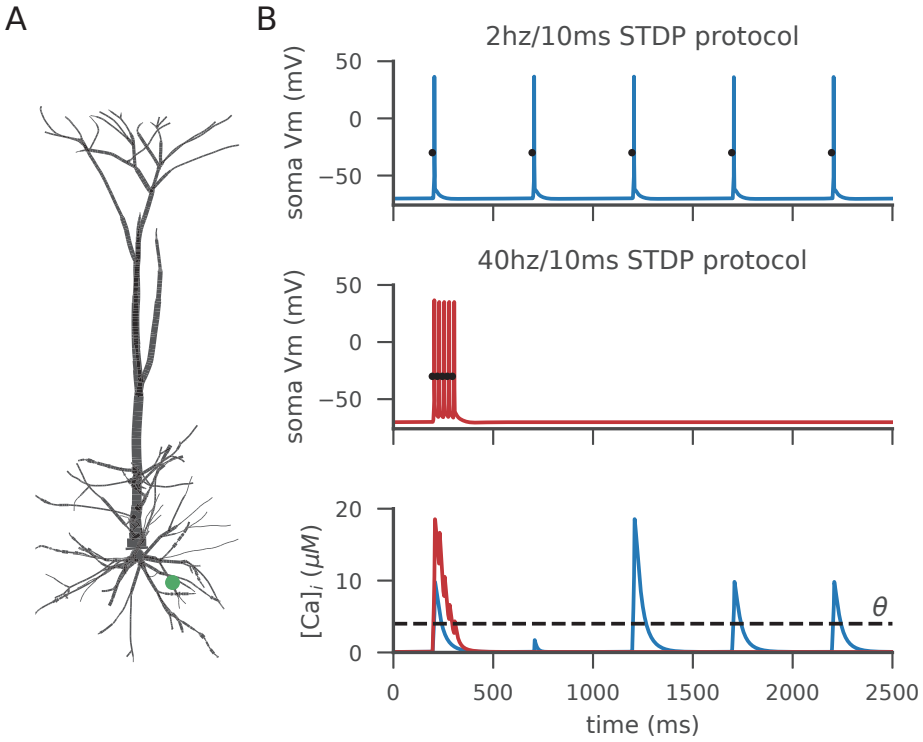


Figure 3.3 – Calcium thresholding problem. A, 2D representation of a typical layer 5 TTPC used in our experiments, green dot marks the position of a synapse used for the analysis in the next panel. B, spiking activity of cell in panel A when stimulated at low frequency (top) and high frequency (middle), black dots mark the spikes of the presynaptic neuron innervating the cell at the analyzed synapse; postsynaptic calcium transients (bottom) for the two stimulation conditions; black dashed line marks a hypothetical threshold θ.

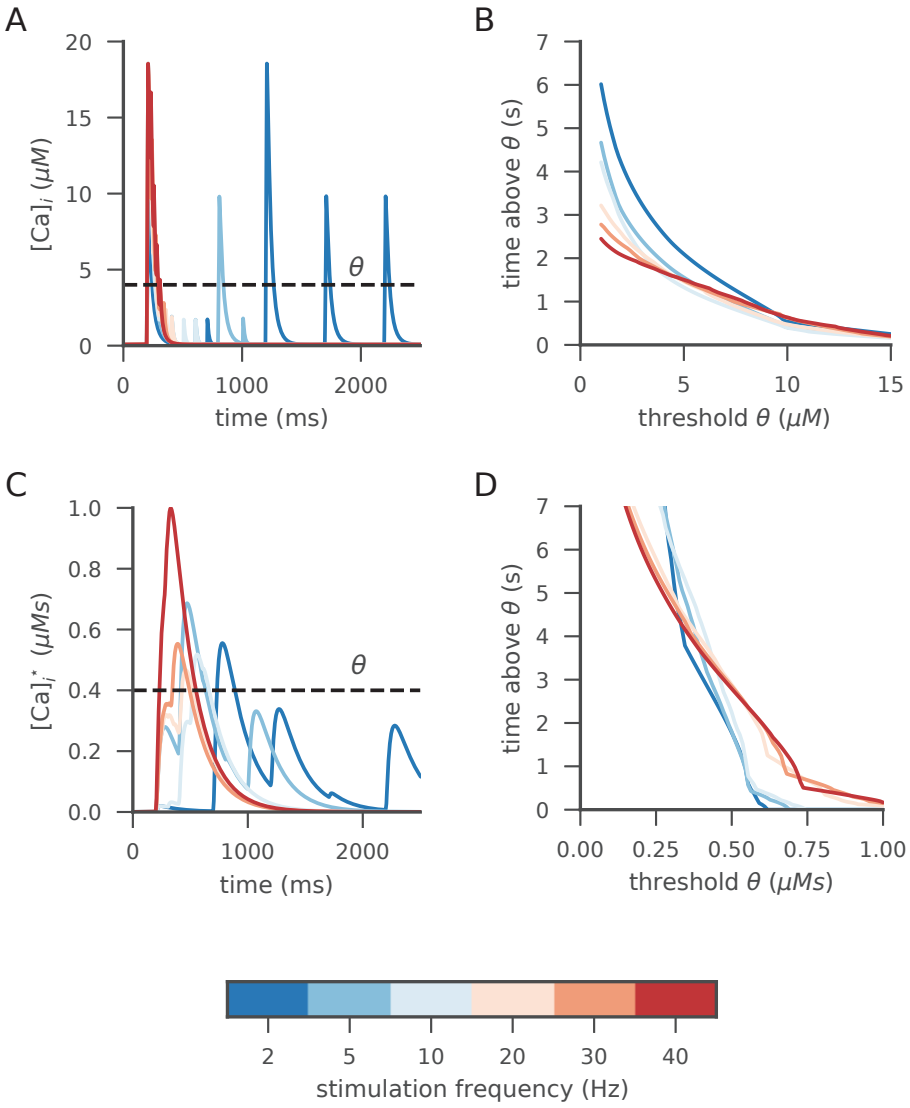


Figure 3.4 – Calcium thresholding solution. A, exemplar calcium transient for a layer 5 TTPC synapse stimulated at various frequencies during a single repetition of a typical STDP inducing protocol; black dashed line marks an hypothetical threshold θ. B, total time spent by postsynaptic calcium above threshold θ during the full STDP inducing protocol. C, same as A, but showing the time course of the calcium integrator instead. D, same as B, but time above thresholds is computed on the calcium integrator instead.

3.1.7 Rewiring

The rewiring model was adapted from Fauth et al. [2015]. In this model the probability of creating a new synapse p_{build} at a potential location is assumed to be constant, while the probability of eliminating an existing one p_{del} is a function of the current synaptic strength, reflecting the experimental findings of [Holtmaat et al., 2005] and Le Bé and Markram [2006], who showed that weak spines/connections are more likely to be eliminated:

$$p_{\text{del}} = p_{\text{build}}^{k} e^{-a^2 w_{ij}^{4/3}} \tag{3.20}$$

where k is a scaling factor controlling the generation to elimination rate; a is the gain of the synaptic weight w_{ij}.

The main modifications with respect to Fauth et al. [2015] were the use of ρ from the GB model instead of the generic weights w and the replacement of Eq. 3.20 with a better behaved logistic function. The latter has the advantage of being monotonic and defined for negative values, allowing a simpler handling of some limit cases (i.e. extremely weak synapses where the efficacy variable ρ is transiently smaller than zero):

$$p_{\text{del}} = \frac{L}{1 + e^{-k(\rho - \rho^{\star})}} \tag{3.21}$$

where L is the maximum value, that is, the elimination probability of a fully depressed synapse; k controls is the steepness of the curve; ρ and $\rho_{\star}$ have their usual meaning (see Sec. 3.1.6).

Parametrization of the rewiring model is based on both experimental evidence and theoretical considerations on the nature of potential synaptic contacts:

- **Total number of synapses decreases with age** already in a time window of 24 hours (from P16 to P17), about 30% of the imaged spines disappeared while only 25% new spines were created [Holtmaat et al., 2005].
- **Newly formed connections are weak** and have low release probability [Le Bé and Markram, 2006].
- **Weak connections have an higher chance of being eliminated**, although newly formed ones seem to be protected from immediate pruning [Le Bé and Markram, 2006].
- **Potential synapses greatly outnumber functional ones** as noticed by Stepanyants et al. [2002].

Based on these observations, we assumed a transition schema for rewiring as the one in Fig. 3.5. A synapse can only be in one of the two structural states: functional or potential [Reimann et al., 2015]. We further distinguish between potentiated and depressed synapses to account for the different elimination probability. A transition from state i to state j is a stochastic event, happening with probability $p_{i,j}$.

Fitting the rewiring model corresponds then to estimate the probability of the nine possible state transitions. To solve the problem, we first considered all the constraints we

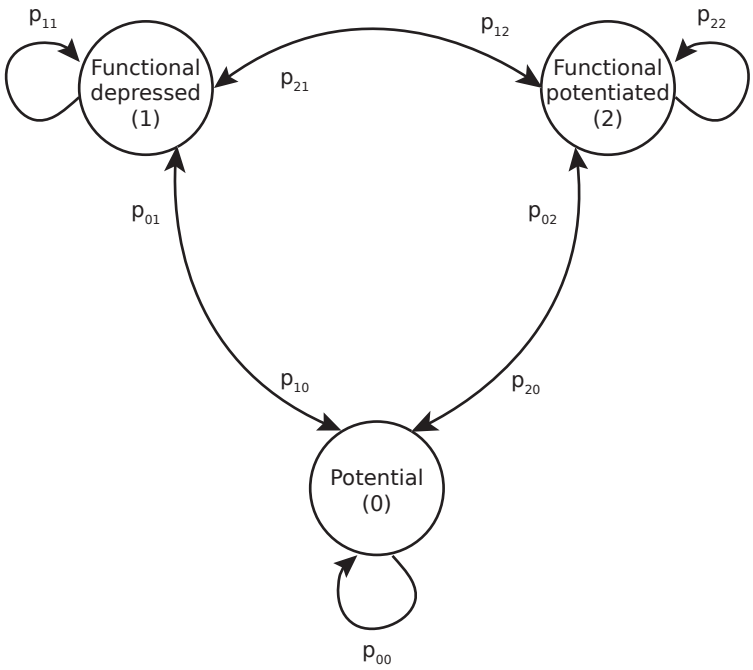


Figure 3.5 – Rewiring model. We consider three possible synaptic states $[0, 1, 2]$, namely *potential*, *functional depressed* and *functional potentiated*. Transition probabilities are indicated by p_{ij} labels.

could derive from experimental evidence and from the BBP's connectivity rules [Reimann et al., 2015]. The probability of a new spine appearing from P16 to P17 is on average 23% [Holtmaat et al., 2005]. Furthermore, potential synapses exceed functional synapses by a factor 16 [Markram et al., 2015, Reimann et al., 2015] (see Fig. 3.6). In mathematical terms:

$$p_{01} + p_{02} = \frac{0.23}{16} \approxeq 0.014 \tag{3.22}$$

The probability of a spine disappearing from P16 to P17 is on average 0.32 [Holtmaat et al., 2005]. In our circuit 50% of the synapses are potentiated and 50% are depressed (see Sec. 3.2.1). Leading to:

$$0.5p_{10} + 0.5p_{20} = 0.32 \tag{3.23}$$

Newly formed synapses are small [Holtmaat et al., 2005] and whole new connections are weak [Le Bé and Markram, 2006]. For this reason the probability to immediately create a very strong synapse is negligible:

$$p_{02} = 0 \tag{3.24}$$

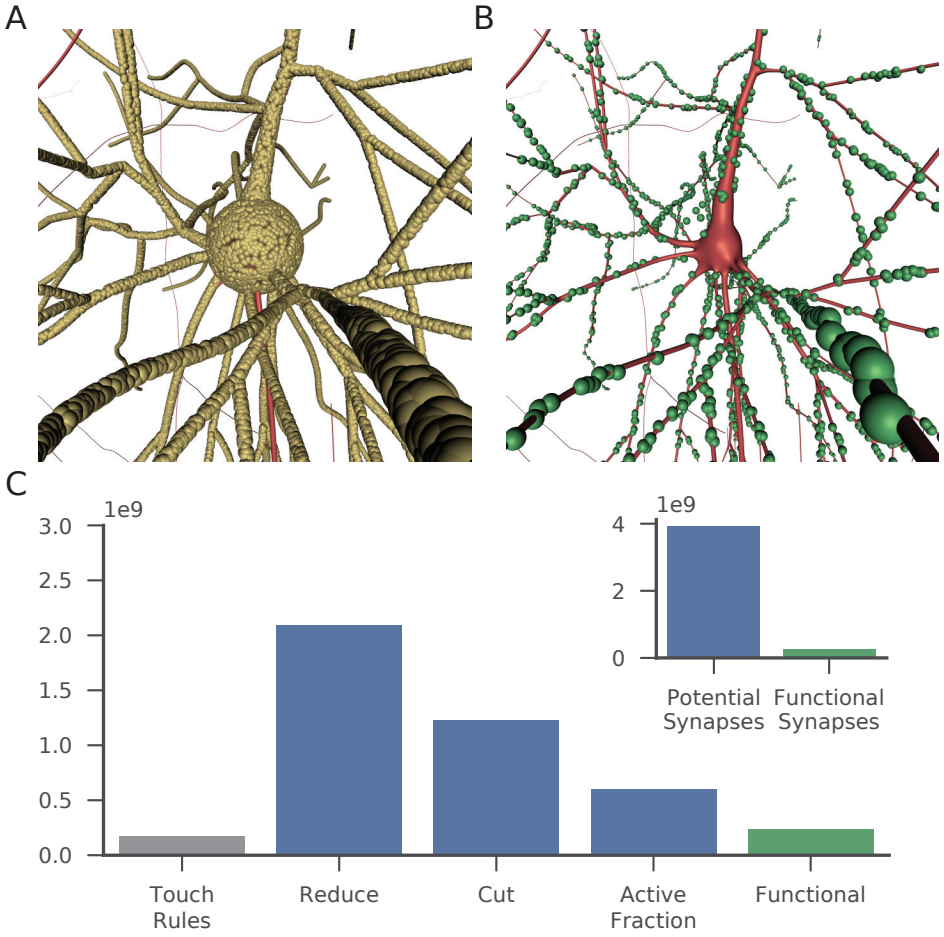


Figure 3.6 – Potential synaptic locations in the BBP's reconstruction. A, all potential synaptic locations around the soma of a single layer 5 TTPC, as determined by the "touch detection" algorithm [Markram et al., 2015]; yellow spheres represent potential synapses, the round soma is an artifact of the detection algorithm, using a fixed radius rather than the actual morphology for computational reasons. B, functional synapses on the same cell in A; the pruning process is carried on by a multi-step filtering algorithm enforcing several biological constraints (i.e. bouton density and number of synapses per connection) [Reimann et al., 2015]. C, excitatory-to-excitatory synapses pruned at every step of the algorithm; blue bars represent the pool of potential synapses used in this work, green the initial count of functional synapses; the inset compares the overall number of potential synapses against functional ones.

The turnover rate for small spines is much larger than the one of big spines [Holtmaat et al., 2005]. Although spine size and synaptic "strength" are well correlated, it is very difficult to quantitatively map synapse properties to elimination rate in our datasets. For this reason, we simply assumed an order of magnitude difference between the elimination

probability of potentiated and depressed synapses:

$$p_{10} - 10p_{20} = 0 \tag{3.25}$$

We complemented the four equations above with other two, based on theoretical considerations on the potentiation/depression processes. In particular, we assumed that the probability of a depressed synapse to remain depressed or to potentiate are identical. Indeed we don't see why a newly formed synapse, depressed by definition as explained before, would not potentiate in the 24h observation window. We than added the equation:

$$p_{11} - p_{12} = 0 \tag{3.26}$$

Furthermore, we want the ratio of depressed and potentiated synapses to be maintained, until further evidence demonstrate a different trend. We enforce this constraint by imposing the net flux of the depressed state to be identical to the one of the potentiated state:

$$p_{01} - p_{02} - p_{10} + p_{11} - 2p_{12} + p_{20} + 2p_{21} - p_{22} = 0 \tag{3.27}$$

Finally, we included three constraints to preserve the consistency of states and transitions:

$$p_{00} + p_{01} + p_{02} = 1 \tag{3.28}$$

$$p_{10} + p_{11} + p_{12} = 1 \tag{3.29}$$

$$p_{20} + p_{21} + p_{22} = 1 \tag{3.30}$$

The resulting linear system can be easily solved analytically. The final values for the $p_{i,j}$ parameters are:

$$\begin{bmatrix} p_{00} \\ p_{01} \\ p_{02} \\ p_{10} \\ p_{11} \\ p_{12} \\ p_{20} \\ p_{21} \\ p_{22} \end{bmatrix} \approxeq \begin{bmatrix} 0.986 \\ 0.014 \\ 0 \\ 0.582 \\ 0.209 \\ 0.209 \\ 0.058 \\ 0.553 \\ 0.388 \end{bmatrix} \tag{3.31}$$

Fitting the logistic function in Eq. 3.21 to $p_{10} = 0.582$ and $p_{20} = 0.058$ leads us to the final parameters for the elimination function: $L = 0.640$ and $k = -4.605$, completing the rewiring model (Fig. 3.7A). We then re-fitted the curve proposed by [Fauth et al., 2015] (see Eq.3.20) and compared it with our formulation (Fig. 3.7B). We noticed that our model shows a more marked saturation for small values of synaptic efficacy than the one from [Fauth et al., 2015]. Furthermore, as mentioned before, the logistic formulation here used (see Eq.3.21) has the advantage of being defined and well behaved for efficacy values outside the (0, 1) range. In addition, we considered an alternative formulation of Eq. 3.20, used by Fauth et al. [2015] to test the sensitivity of their results to the chosen

parameters. They found that indeed their conclusions hold also for the alternative formulation [Fauth et al., 2015]. As this second version of the elimination curve is very similar to our sigmoid, we are confident that it was a valid replacement. Synapse

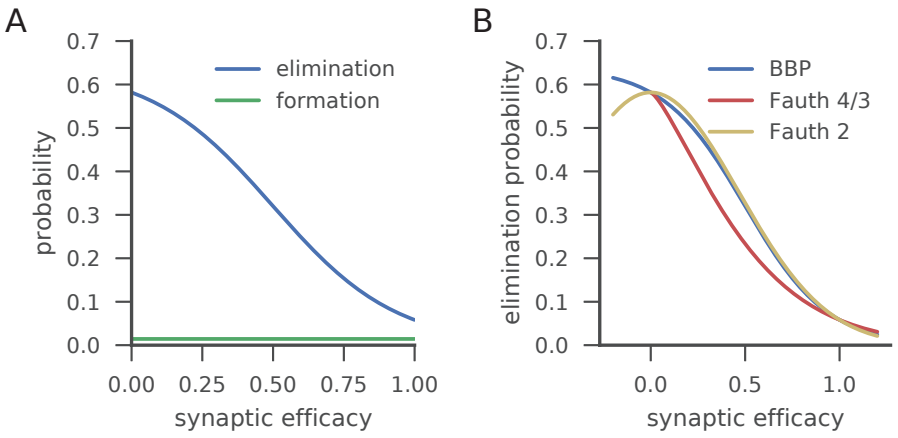


Figure 3.7 – Rewiring fit. A, synapse formation and elimination probabilities; synapses emerge at a constant rate, while their elimination probability depends on synaptic efficacy. B, comparison of the elimination curve used in this work (BBP, see Eq. 3.21) with the one used in Fauth et al. [2015] (Fauth 4/3, see Eq. 3.20); an alternative formulation where the w_{ij} exponent in Eq. 3.20 is changed from 4/3 to 2 is also shown (Fauth 2) [Fauth et al., 2015]; the x axis is purposely include points smaller than 0 and greater than 1 to show the behavior of the three functions for limit and pathological cases.

formation and elimination were then implemented as an Poisson processes of parameter $p_{\text{build}} = p_{01}$ and p_{del} respectively (see 3.4.3 for technical details on the implementation of the elimination process).

3.2 Fitting the long-term plasticity model

Observing the activity of single synapses undergoing functional plastic changes is nowadays generally not possible, with notable exceptions [Nevian and Sakmann, 2006]. For this reason, the vast majority of available datasets contain only information at the level of whole (multi-synaptic) connections [Markram et al., 1997b, Sjöström et al., 2001]. This lack complicates quite dramatically the task of constraining models of synaptic plasticity to experimental data. To overcome the problem, we designed a fitting procedure capable of extracting a suitable parameterization for single synapses from whole connection data.

In brief, a pool of in silico connections between layer 5 TTPC is randomly chosen from our circuit reconstruction [Markram et al., 2015]. The connections are then stimulated by pairing the activity of the pre- and post-synaptic cells to trigger functional synaptic plasticity, as in standard in vitro STDP experiments [Markram et al., 1997b, Sjöström

et al., 2001]. A genetic algorithm identifies the best set of synaptic parameters by comparing the in silico results with the in vitro dataset in Markram et al. [1997b]. The obtained parameter set is then generalized to all other synaptic pathways.

In the following sections we describe the details of the fitting procedure. We start by stating the optimization problem in Sec. 3.2.1 and how neuron pairs were selected in Sec. 3.2.2. Then we characterize the actual fitting method in Sec. 3.2.3. Finally, the generalization steps are discussed in Sec. 3.2.4 and 3.2.5.

3.2.1 Problem statement and initial conditions

To fit the long-term plasticity model to experimental data we need to find appropriate values for several parameters:

$\boldsymbol{\tau}$	time constant of the synaptic efficacy;
$\boldsymbol{\theta_d}$, $\boldsymbol{\theta_p}$	threshold for depression and potentiation respectively;
$\boldsymbol{\gamma_d}$, $\boldsymbol{\gamma_p}$	rate for depression and potentiation respectively;
$\boldsymbol{\tau_{\mathsf{change}}}$	time constant of long-term release probability changes;
$\boldsymbol{U_{\mathsf{SE}}^{(d)}}$, $\boldsymbol{U_{\mathsf{SE}}^{(p)}}$	absolute release probability in the depressed and potentiated state respectively;
$\boldsymbol{\beta_0}$	initial ratio of depressed synapses.

We decided to fix some parameters and to limit the search space of others to our best guess to simplify the fitting procedure (see table 3.1).

Parameter	Value	Unit
τ	70	s
θ_d	(0.25, 0.5)	µM s
θ_p	(0.25, 0.5)	µM s
γ_d	(1, 350)	scalar
γ_p	(1, 350)	scalar
τ_{change}	100	s
$U_{\text{SE}}^{(d)}$	0.2	scalar
$U_{\text{SE}}^{(p)}$	0.8	scalar
β_0	0.5	scalar

Table 3.1 – Parameters of the long term plasticity model. Values between parentheses represent search boundaries (lower, upper), while the others are fixed parameters.

The two time constants τ and τ_{change} were estimated from preliminary fitting attempts and are compatible with previously used parameter sets (see Supplementary Information in Graupner and Brunel [2012]). The initial ratio of depressed synapses β_0 was inherited from Graupner and Brunel [2012], who successfully used this value to fit another

neocortical STDP dataset by Sjöström et al. [2001]. The release probability of the depressed state $U_{\mathrm{SE}}^{(d)}$ and the one of the potentiated state $U_{\mathrm{SE}}^{(p)}$ were chosen to satisfy both constraints on the maximum potentiation and depression attainable and to preserve the mean release probability of the pathway. Specifically:

Mean release probability constraint as in Markram et al. [2015]

$$U_{\mathrm{SE}}^{(\mathrm{initial})} = \beta_0 U_{\mathrm{SE}}^{(d)} + (1-\beta_0) U_{\mathrm{SE}}^{(p)} = 0.5 \tag{3.32}$$

Maximum depression as in Markram et al. [1997b]

$$\Delta\mathrm{EPSP}_d = \frac{U_{\mathrm{SE}}^{(d)}}{U_{\mathrm{SE}}^{(\mathrm{initial})}} \leq 0.8 \tag{3.33}$$

Maximum potentiation as in Markram et al. [1997b]

$$\Delta\mathrm{EPSP}_p = \frac{U_{\mathrm{SE}}^{(p)}}{U_{\mathrm{SE}}^{(\mathrm{initial})}} \geq 1.5 \tag{3.34}$$

All previous formulas were derived from results in Graupner and Brunel [2012] and led us to the following set of constraints for $U_{\mathrm{SE}}^{(d)}$ and $U_{\mathrm{SE}}^{(p)}$:

$$\begin{cases} \beta_0 U_{\mathrm{SE}}^{(d)} + (1-\beta_0) U_{\mathrm{SE}}^{(p)} = 0.5 \\ 0 \leq U_{\mathrm{SE}}^{(d)} \leq 0.4 \\ 0.75 \leq U_{\mathrm{SE}}^{(p)} \leq 1 \end{cases} \tag{3.35}$$

The solution of this system of equalities and inequalities is a linear set, graphically shown in Fig. 3.8. We validated the proposed parametrization for $U_{\mathrm{SE}}^{(d)}$ and $U_{\mathrm{SE}}^{(p)}$ by reproducing in silico a typical in vitro connectivity test. Our goal was to verify that the $U_{\mathrm{SE}}^{(\mathrm{initial})}$ extracted at the connection level was still compatible with what reported experimentally [Markram et al., 1997a]. We considered the standard "H20S8" stimulation protocol, as used in our lab to probe connection properties: given a connected pair of neurons, the presynaptic one is forced to spike 8 times at a frequency of 20 Hz; a delayed supplementary spike ($\Delta t = 550$ms) is then delivered to show recovery from depression dynamics; the protocol is repeated 60 times and the average somatic Excitatory Post-Synaptic Potential (EPSP) computed. We selected a pool of 200 in silico connections satisfying the typical geometrical constraints holding in vitro (see Sec. 3.2.2). The synapse model used for this test does not include any forms of long-term plasticity nor VDCC[3]. After simulation, $U_{\mathrm{SE}}^{(\mathrm{initial})}$ was calculated using a Genetic Algorithm (GA) [Van Geit et al.,

[3] Long-term plasticity and the VDCC considered in this work are not likely to affect the results of our test. The former operates at much longer time scales. The latter would have very little influence on the somatic EPSP, as only HVA channels are considered here.

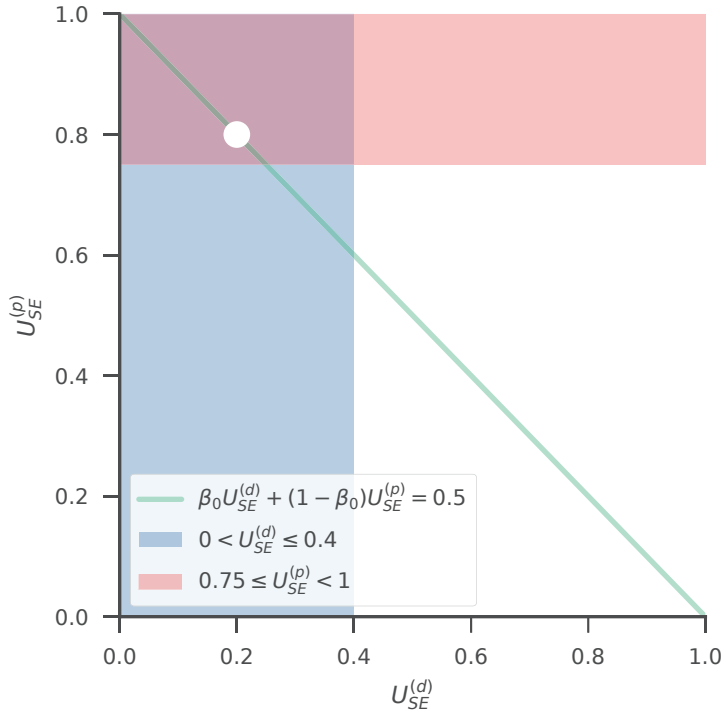


Figure 3.8 – Set of valid solutions for $U_{\mathrm{SE}}^{(d)}$ and $U_{\mathrm{SE}}^{(p)}$. Red, green and blue sets represent data points satisfying the prescribed constraints in Eq. 3.35. The intersection of the three is a line segment, corresponding to the set of valid solutions. White circle shows the chosen value for $U_{\mathrm{SE}}^{(d)}$ and $U_{\mathrm{SE}}^{(p)}$.

2016] to fit the TM model to the somatic traces (see. Fig. 3.9). Our results show that, despite bimodal synapses, the release probability extracted at the connection level is still unimodal. It should be noticed that the mean value we estimated is slightly higher than expected. There is indeed a tendency to overestimate $U_{\mathrm{SE}}^{(\mathrm{initial})}$ for connections composed by a mixture of potentiated and depresses synapses. This issue will be addressed in the next iteration of the synapse model (see Sec. 5.2).

After fixing boundaries and values of several parameters of the long-term plasticity model, we focused on the other synaptic parameters. Our first goal was to obtain a mean model of calcium-dependent plastic changes for layer 5 TTPC connections, neglecting for a moment all the complications due to variability of other synaptic parameters (i.e. peak conductance or release probability). For this reason we fixed them to their mean reported values (see table 3.2 or Ramaswamy et al. [2015]). The variability of transmission parameters was later re-introduced and accounted for in the corresponding generalization step (see Sec. 3.2.4). The fitting problem is then finally reduced to the estimation of just four parameters: the two thresholds θ_d and θ_p, plus the corresponding rates γ_d and γ_p.

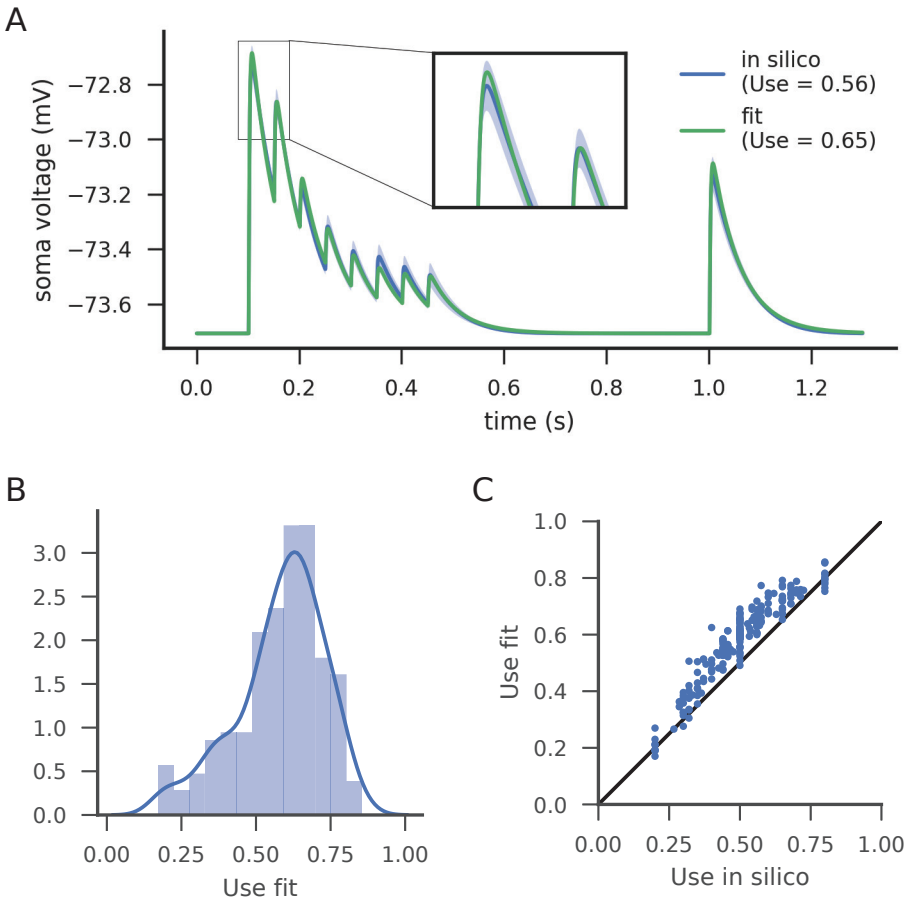


Figure 3.9 – Bimodal $U_{\mathrm{SE}}^{(\mathrm{initial})}$ validation. A, sample connection and corresponding fit; blue line is the mean response of 60 repetitions of the stimulation protocol, filled blue area represents the 95% confidence interval; green line is the fit produced by the algorithm. B, $U_{\mathrm{SE}}^{(\mathrm{initial})}$ distribution as extracted by the fitting algorithm for every connection in the in silico dataset; blue line represents the kernel density estimate. C, prescribed mean $U_{\mathrm{SE}}^{(\mathrm{initial})}$ versus extracted $U_{\mathrm{SE}}^{(\mathrm{initial})}$; the fitting algorithm tends to overestimate $U_{\mathrm{SE}}^{(\mathrm{initial})}$.

Parameter	NMODL	Value	Unit
Peak conductance	gmax	1.5	nS
Conductance decay time constant	tau_d	1.74	ms
Depression time constant	Dep	670	ms
Facilitation time constant	Fac	17	ms
Size of readily releasable vesicle pool	Nrrp	2	scalar

Table 3.2 – Fixed synaptic parameters.

3.2.2 Pair selection

We selected in silico pairs of connected layer 5 TTPC to match the key features of standard STDP experiments in vitro [Markram et al., 1997b, Sjöström et al., 2001]. In particular, we considered two type of constraints, namely electrical and geometrical, next described.

Layer 5 TTPC were first tested for stimulation protocol compliance. That is, each cell was stimulated as in the STDP induction protocols and discarded in case of missing or redundant spikes (see Fig. 3.10)[4]. Cells properly responding to all stimulation protocols were then added to the pool of valid postsynaptic cells. The same constraint was not enforced on presynaptic cells as they are not actually simulated in our fitting procedure. The presynaptic spikes are instead simply delivered to the synapse at the appropriate time. This choice reduce the computational cost of fitting and does not impact on its quality in any way. That is, in our network model, and more generally in NEURON, spikes are anyway simply represented as discrete events delivered to the synapse, located on the postsynaptic neuron. The actual waveform of the spike has no effect on the response of the postsynaptic cell. Axonal delay is accounted for in the spike delivery system and we added an additional 5 ms delay to compensate for the membrane depolarization delay (see Fig. 3.10C).

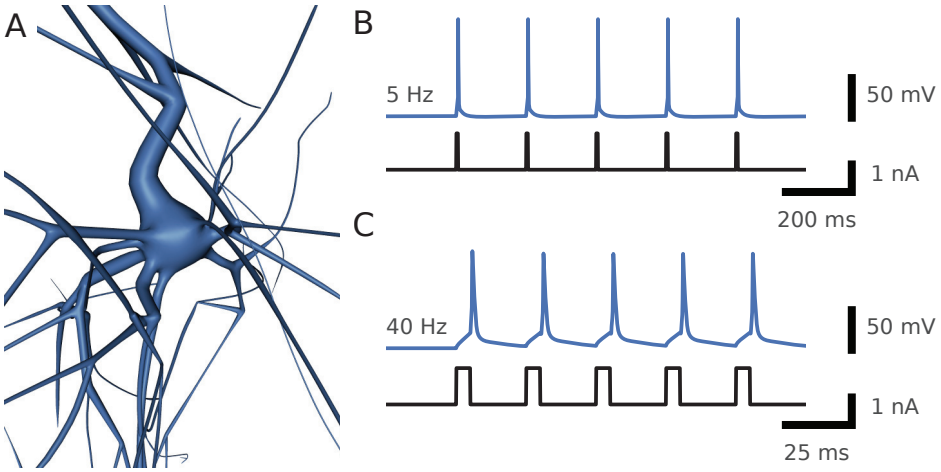


Figure 3.10 – Exemplar postsynaptic cell used for fitting. A, close-up of a typical layer 5 TTPC, selected as a candidate postsynaptic cell. B, spiking activity of cell in A (top) at 5 Hz stimulation (bottom). C, spiking activity of cell in A (top) at 40 Hz stimulation (bottom).

[4] A relaxation of the electrical model for pyramidal cells was required to allow proper spiking at high-frequency stimulation. Specifically, the decay time constant of calcium dynamics at the soma was reduced to eliminate excessive hyper-polarization after a rapid series of spikes (data not shown). Although this modification alters the electrical properties of the cell, its application is confined to model fitting. Furthermore, we would like to remark that the typical STDP stimulation protocols are quite artificial in terms of spiking activity and do not represent a typical network scenario. For this reason, we decided to avoid a proper re-fitting of the electrical model and accepted the relaxation of a single parameter as a valid approximation.

After electrical screening, we focused on the geometrical constraints of typical STDP experiments. Experimentalists tend to patch cells close to each other to increase the chances of finding a synaptic connection, as connection probability decays rapidly with cells distance [Perin et al., 2011]. Furthermore, to simplify the patching process, the selected cells are often on the same focal plane (same depth in the tissue). To reproduce these technical biases, each pair was selected by extracting a random (postsynaptic) cell A from the pool of candidate previously defined and building a bounding box of $100\,\mu m \times 100\,\mu m \times 10\,\mu m$ (x, y, z coordinates) around it. All other (presynaptic) cells in the bounding box were identified and tested for connection with cell A. The first match, if any, was chosen to complete the pair. Following these constraints, we produced two sets of 50 connections each, namely the *training set* and the *testing set* (Fig. 3.11). The

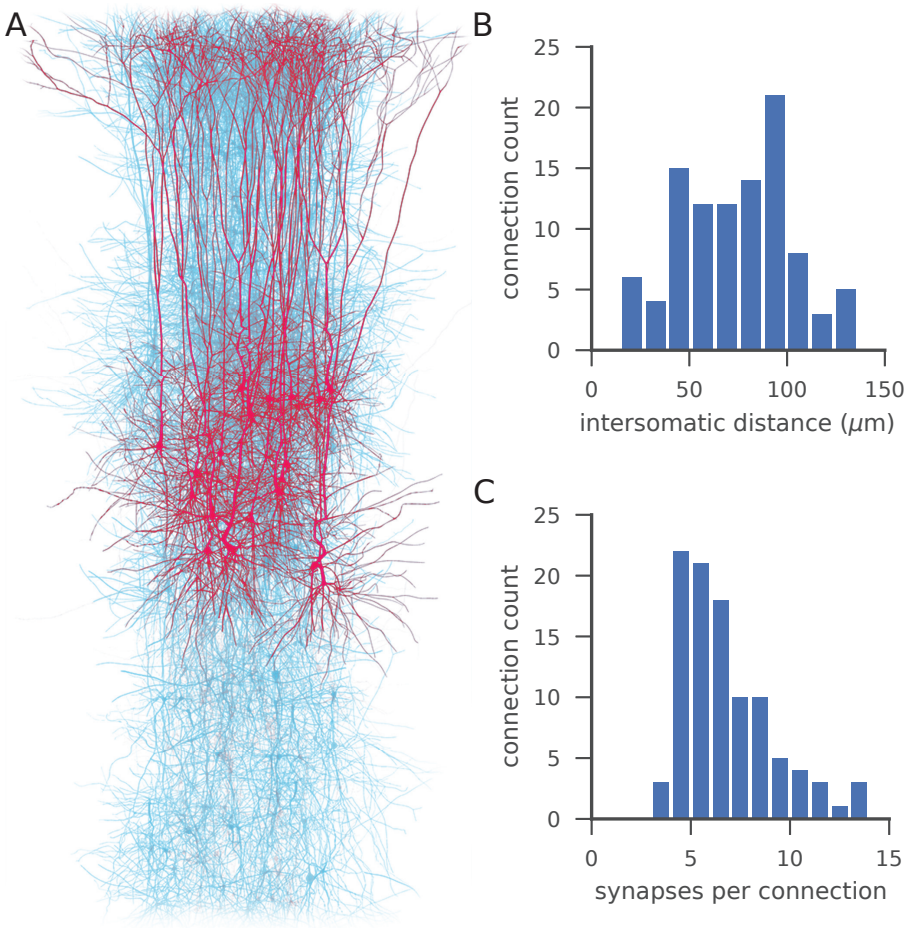


Figure 3.11 – Geometrical constraints on cell selection. A, exemplar layer 5 TTPC cells used for fitting. B, intersomatic distance distribution for all pairs in the training and testing set. C, number of synapses per connection distribution for same pairs in B.

former was used for parameter fitting via Multi-Objective Optimization (MOO), as later

discussed, while the latter was used to asses the generalization power of the solution.

3.2.3 Parameter fitting

To fit the extended GB model we considered in vitro procedures in Markram et al. [1997b], Sjöström et al. [2001] and reproduced them in silico. We focused on three stimulation protocols[5]and their outcome:

H10minus an LTD-inducing protocol. Both the pre- and post-synaptic cells are forced to spike at a frequency of 10 Hz, but presynaptic current injections are delayed by 10 ms with respect to the postsynaptic ones.

H2plus a baseline protocol. Both the pre- and post-synaptic cells are forced to spike at a frequency of 2 Hz, but presynaptic current injections are anticipated by 10 ms with respect to the postsynaptic ones.

H2plus a mild LTP-inducing protocol. Both the pre- and post-synaptic cells are forced to spike at a frequency of 10 Hz, but presynaptic current injections are anticipated by 10 ms with respect to the postsynaptic ones.

The effects of the three stimulation protocols on a typical synapse are shown in Fig. 3.12.

A typical STDP protocol consists of three phases: (1) assessment of initial Post-Synaptic Potential (PSP) amplitude, (2) plasticity induction, (3) assessment of final PSP amplitude. Assessing PSP amplitude is particularly time consuming because it requires many trials (usually more than 30). Furthermore, in case of after-protocol PSP assessment (i.e. phase 3 above), the effects need to be evaluated after a sufficiently long time (up to 30 min) to be sure that the any change could be really considered *long-term*. Unfortunately, simulating such a long protocol is an extremely expensive task. For this reason, parameter fitting could not be performed by exactly reproducing all the steps of an STDP experiment. However, given that in our model the mean ΔEPSP only depends on the ratio of depressed synapses β:

$$\Delta\text{EPSP} = \frac{\beta U_{\text{SE}}^{(d)} + (1-\beta)U_{\text{SE}}^{(p)}}{\beta_0 U_{\text{SE}}^{(d)} + (1-\beta_0)U_{\text{SE}}^{(p)}} = \frac{\beta U_{\text{SE}}^{(d)} + (1-\beta)U_{\text{SE}}^{(p)}}{\overline{U_{\text{SE}}}} \tag{3.36}$$

and that θ_d and θ_p thresholds are likely to be larger than calcium transients during connectivity test, where the postsynaptic cell is not spiking, we could fit the model by simulating only the induction phase and computing β before and after it (β-strategy).

We used a multi-objective GA Van Geit et al. [2016] to find model parameters best matching in vitro data (see 3.13). The final solution found by the GA was then validated by fully simulating all available STDP protocols (including connectivity test) on both the training and the testing set. Our results show not only that the β-strategy was a valid approach but also that the model well capture the typical LTP frequency-dependence and the switch from LTD to LTP at high-frequency [Sjöström et al., 2001].

[5] To standardize the induction procedure and to comply with the methods in more recent studies [Sjöström et al., 2001], we fixed abs(Δt) = 10 ms, $n = 15$ and $T = 10$ s for all protocols.

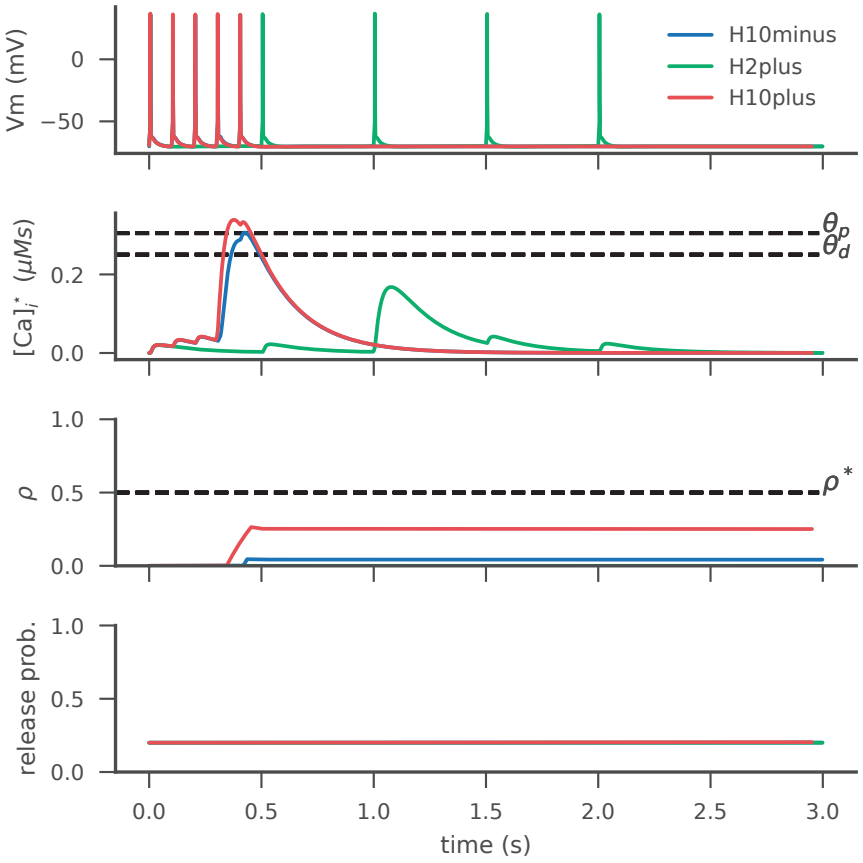


Figure 3.12 – Effects of STDP protocols on a single synapse in silico. From top to bottom: soma voltage of postsynaptic cell during a single repetition of the protocol, presynaptic spikes not shown for clarity; effective calcium concentration at the synapse under analysis, an appropriate choice of the thresholds allows to correctly recognize the stimulation protocols; the LTP-inducing protocol triggers an increase of synaptic efficacy, while the other protocols do not; release probability is not immediately affected by changes of synaptic efficacy.

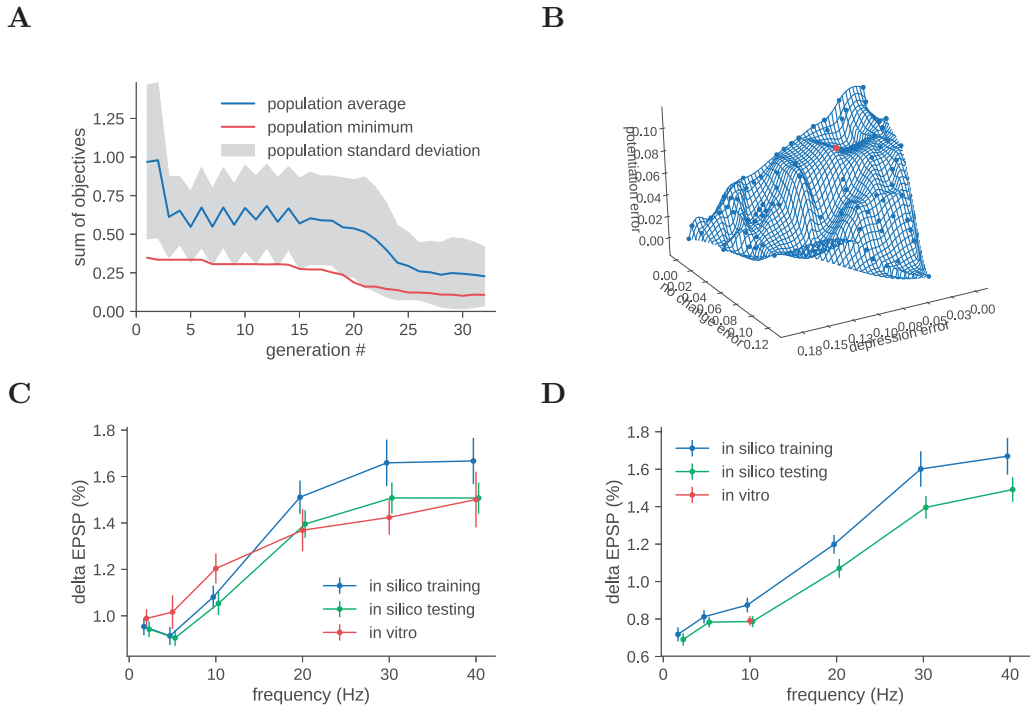


Figure 3.13 – LTP/LTD model fitting results. A, GA error evolution during fitting; blue line represents the mean error for all candidate solutions in the population; gray area is the standard deviation of the population error; red line the error of the best solution in the population. B, Pareto front obtained after fitting, marked in red the overall best solution. C, performance of the best solution on the training (blue) and testing (green) sets, compared to in vitro data (red), for the LTP inducing protocols; error bars represent SEM. D, performance of the best solution on the training (blue) and testing (green) sets, compared to in vitro data (red), for the LTD inducing protocols; error bars represent SEM.

3.2.4 Pathway generalization

As previously mentioned, in the model fitting step we intentionally neglected the variability of synaptic parameter, such as the density of glutamate receptors (Section 3.2.1).

Synapses can be located on large or small spines and bear a large or small population of calcium permeable channels. These two factors are, to first approximation, the main drivers of postsynaptic calcium concentration. Spine volume and Postsynaptic Density (PSD) area are strongly correlated [Arellano et al., 2007]. This means that generally "strong" ("weak") synapses are located on large (small) spines. This fact provide support for an intrinsic "calcium normalization" at the level of individual spines. We explicitly modeled this relationship by correlating spine volume and conductance at each synapse, as described in Section 3.1.4. However, this basic "calcium normalization" was not sufficient to reproduce the in vitro STDP results (data not shown).

To further investigate the relationship between synaptic parameters and calcium concentration, we simulated a long connectivity test protocol (40 minutes) and recorded the peak of the effective calcium concentration. We discovered that this value is well correlated with gmax/vol and the relationship can be safely modeled with a linear function, as shown in Fig. 3.14A. We then hypothesized that conditioning the LTP and LTD thresholds to gmax/vol could allow us to reproduce the in vitro results, such as

$$\theta_x = m_x \frac{\text{gmax}}{\text{vol}} + b_x \tag{3.37}$$

in particular we imposed the following two constraints:

- $m_x = m_{\text{peak}}$, same slope of peak and gmax/vol linear function
- $\theta_x(\text{defult}) = m_x\text{default} + b$, as fitted in step 2 (Sec. 3.2.3)

which in turn lead to the following:

$$\begin{aligned} \theta_d &= 0.015\frac{\text{gmax}}{\text{vol}} - 0.003 \\ \theta_p &= 0.015\frac{\text{gmax}}{\text{vol}} + 0.051 \end{aligned} \tag{3.38}$$

Fig. 3.14B shows how the new threshold compare to the calcium peaks during connectivity test. This generalization strategy finally allowed us to reproduce the in vitro STDP results and recover the same performance of the mean model fitted in Sec. 3.2.3, as shown in Fig. 3.14C and Fig. 3.14D. After scaling all the thresholds using the conductance-volume ratio, we can reproduce in silico a classical STDP experiment as in Markram et al. [1997b] (Fig. 3.15).

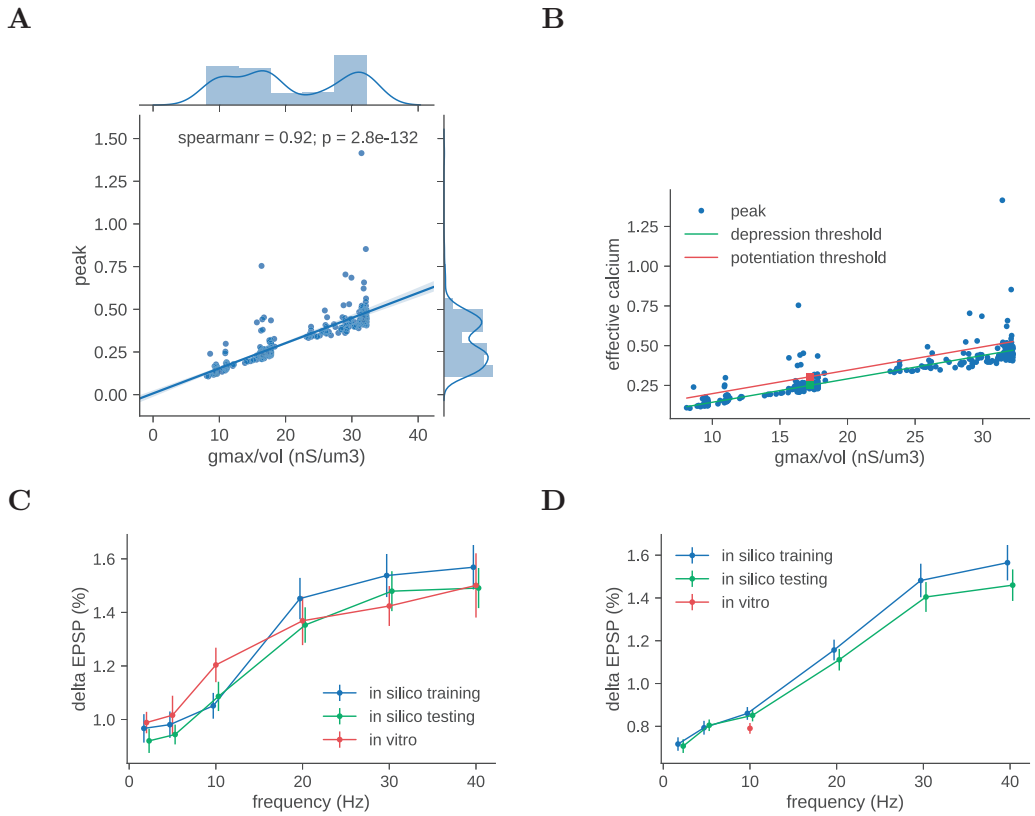


Figure 3.14 – Pathway generalization results. A, correlation between peaks of the effective calcium transients during presynaptic-only stimulation and the conductance-volume ratio; blue line represent a linear fit. B, threshold scaling as a function of the conductance-volume ratio; red and green square represent the fitted potentiation and depression threshold respectively. C, performance of the best solution after threshold scaling on the training (blue) and testing (green) sets, compared to in vitro data (red), for the LTP inducing protocols; error bars represent SEM. C, performance of the best solution after threshold scaling on the training (blue) and testing (green) sets, compared to in vitro data (red), for the LTD inducing protocols; error bars represent SEM.

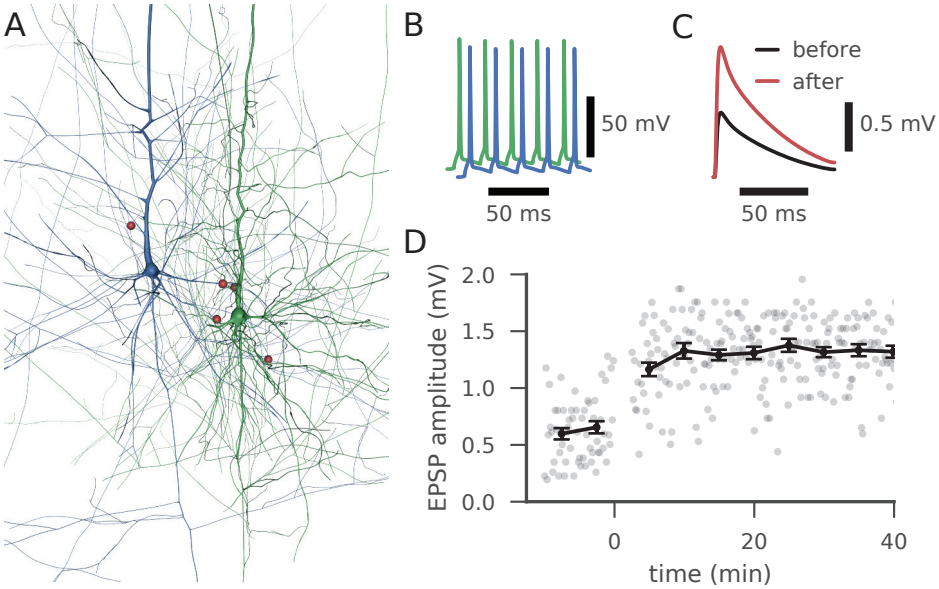


Figure 3.15 – Classical STDP experiment in silico. A, pair of layer 5 TTPC used for the in silico experiment; presynaptic cell in blue, postsynaptic cell in green, synapses in red. B, spiking activity of the pre- and postsynaptic cell during stimulation at 40 Hz; the green trace has only illustration purposes, as the presynaptic cell is not simulated in our experiments. C, mean EPSP amplitude before (black) and after (red) the pairing. D, evolution of EPSP amplitude over time; gray dots represent individual trials, black dots the mean over 5 minutes of the individual trials; error bars represent SEM.

3.2.5 Circuit generalization

The final step towards a comprehensive model of calcium-based plastic changes is the generalization to arbitrary pathways. Unfortunately the experimental validation of this step is quite problematic, given that only a few pathways have been experimentally characterized.

We then assumed the main discriminants between pathways, with respect to synaptic plasticity, to be the innervation profile and the morphology types rather than a different set of plastic mechanisms. We confirmed the validity of this assumption by reproducing an in vitro experiment on soma distance-dependent plasticity, as later shown in Section 3.3.1. For this reason, a minimal variation of our fitted solution, accounting only for the differences in synaptic physiology, would be sufficient to parametrize any uncharacterized pathways.

The generalization procedure computes β, Use_d and Use_p for all pathways, knowing only the mean release probability $\overline{\mathrm{Use}}$. We used the Sequential Least SQuares Programming (SLSQP) method [Kraft, 1988], as implemented in the Scipy package, to find a suitable solution to the minimization problem:

$$\underset{\beta,\mathrm{Use}_d,\mathrm{Use}_p}{\text{minimize}}\ \overline{\mathrm{Use}} - [\beta\mathrm{Use}_d + (1-\beta)\mathrm{Use}_p] \tag{3.39}$$

The solution to this problem is expected to have several local minima. In the absence of other constraints, we imposed the same β, Use_d and Use_p used for layer 5 TTPC, as the initial point for the optimization.

3.3 Validation and predictions

We previously showed how the long-term plasticity model was fitted to available data and how it was then generalized to parametrize a whole neocortical circuit (see Sec. 3.2). We now focus on the validation of this strategy and on the predictions that can be made about uncharacterized pathways.

Validating our generalization strategy is a difficult task, mostly because of the sparsity of experimental data. In Section 3.3.1 we prove the quality of our parameterization by reproducing in silico the results on distance-dependent synaptic plasticity by Sjöström and Häusser [2006]. The idea of distance-dependent plasticity can be summarized as follows: the direction and magnitude of plastic changes induced by a given stimulation protocol depend on both the protocol itself and the innervation pattern of the connection. By showing that distance-dependent plasticity is a consequence of our parametrization, even though the dataset by Sjöström and Häusser [2006] was never considered in the fitting procedure, we provide direct evidence of the validity of our approach. Furthermore, we show how synaptic dynamics, neuronal morphologies and innervation patterns jointly determine the nature of plastic changes, demonstrating once more the need of such a level of modeling detail to even consider the study of higher brain functions, such as learning.

In Section 3.3.2 we extend our STDP analysis to totally uncharacterized pathways. In particular, we simulated the outcome of typical STDP protocols on every possible excitatory to excitatory pathway in our circuit. Such a dataset do not exist in experimental literature and contains interesting indications on the plastic *regimes* observable in a network. Our goal is to show how the proposed model and parametrization can be used to obtain insights on network functioning and to guide future in vitro and in vivo experiments.

3.3.1 Distance-dependent synaptic plasticity

It has been shown that a standard 50 Hz STDP protocol, triggering potentiation of layer 5 TTPC connections, results in depression of distal synapses, i.e. layer 2/3 to layer 5 pyramidal connections [Sjöström and Häusser, 2006]. This finding confirms the hypothesis that excitatory pathways might exhibit different forms of synaptic plasticity as the result of different innervation patterns rather than different plasticity rules.

To test this hypothesis and validate our fitting generalization, we reproduced in silico the experiments in Sjöström and Häusser [2006]. We randomly selected 50 connections between layer 5 TTPC and 50 connections between layer 2/3 pyramidal cells (PCs). To maximize the mean synaptic distance span we searched for cells in the half-top part of layer 2/3 and the half-bottom part of layer 5. Plastic changes were triggered as in previously described [Markram et al., 1997b, Sjöström et al., 2001, Sjöström and Häusser, 2006].

Our results confirm the LTD switch of distal connections, while the proximal ones still exhibited LTP (Fig. 3.16 and Fig. 3.17). Furthermore, we extended our analysis to single synapses and observed a clear transition between LTP and LTD as a function of synaptic distance from soma (Fig. 3.18).

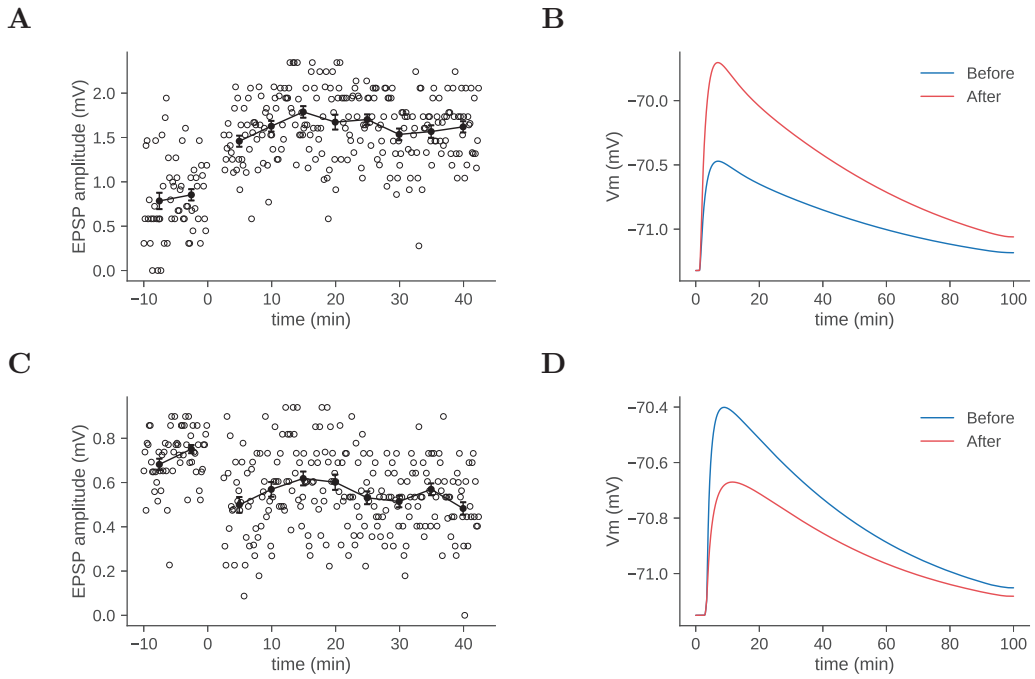


Figure 3.16 – Example of distance-dependent synaptic plasticity. A, evolution of EPSP amplitude over time for a layer 5 TTPC to layer 5 TTPC connection, stimulated at 50 Hz; empty circles represent individual trials, filled circles the mean over 5 minutes od the individual trials; error bars represent SEM. B, mean EPSP amplitude before (blue) and after (red) the pairing for the connection in A. C, evolution of EPSP amplitude over time for a layer 2/3 PC to layer 5 TTPC connection, stimulated at 50 Hz; empty circles represent individual trials, filled circles the mean over 5 minutes od the individual trials; error bars represent SEM. D, mean EPSP amplitude before (blue) and after (red) the pairing for the connection in C.

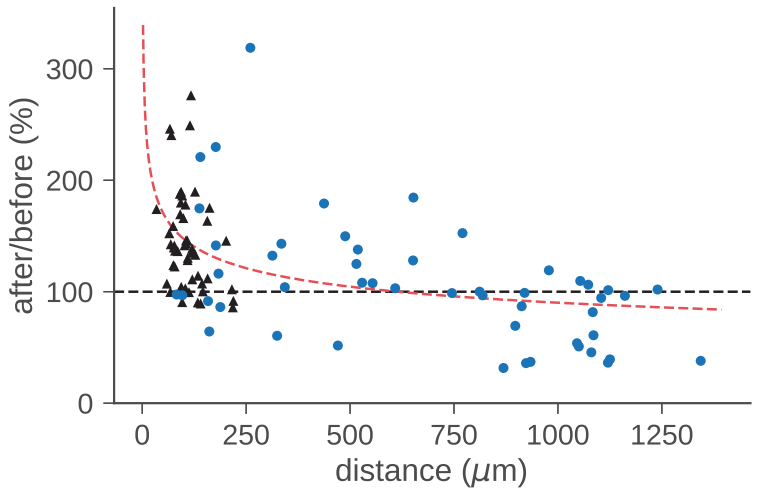


Figure 3.17 – Summary of distance-dependent synaptic plasticity. Potentiation switches to depression for distally connected cells. Triangles show layer 5 TTPC to layer 5 TTPC connections, filled circles layer 23 PC to layer 5 TTPC connections; dashed red line is a polynomial fit to the data.

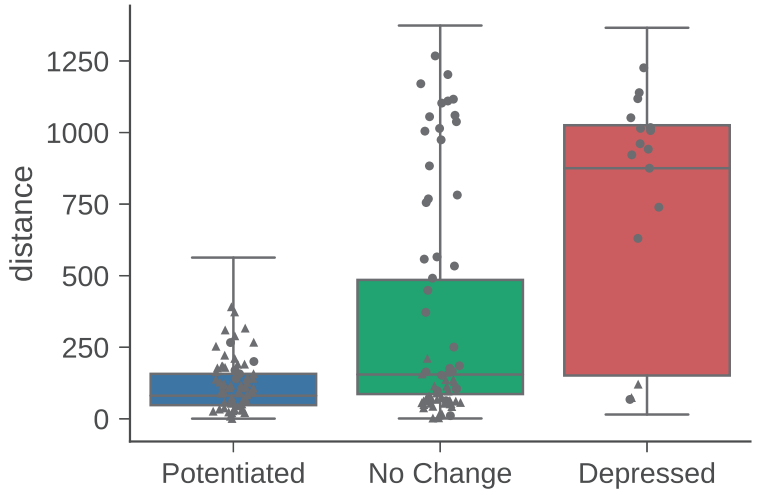


Figure 3.18 – Distance-dependent synaptic changes. Distal synapses are more prone to depression and almost never potentiate. Single data points from a random sample of 150 synapses; layer 5 TTPC to layer 5 TTPC connections represented as triangles; layer 2/3 PC to layer 5 TTPC represented as circles.

3.3.2 STDP map

A systematic characterization of STDP in the neocortex is nowadays missing. Long-term synaptic plasticity has been experimentally studied only at layer 5 TTPC connections and a few other neocortical pathways [Markram et al., 1997b, Sjöström et al., 2001, Froemke et al., 2005, Sjöström and Häusser, 2006] [6], due to the many technical challenges:

- Connections are rare or hard to find for most pathways, especially if the pre- and post-synaptic cells belong to different cortical layers;
- Experiments are inherently long and have high failure rate;
- (Sadly enough) only a full characterization of all pathways or the discovery of a computationally relevant difference between some of them would be worth the effort.

However, it is known that the direction and magnitude of synaptic changes induced by paired neural activity profoundly depend on the morphological features of the neurons and the innervation patterns. For this reason, we could expect great variability of STDP dynamics across pathways (see also Sec. 3.3.1).

In this section we present our attempt to fill the gap by generating a simulated *STDP map*. The map not only provides a comprehensive view of STDP in the neocortex, but also it could be used as a reference to guide future experiments. We simulated the outcome of three STDP protocols (2, 10 and 50 Hz) for all 169 excitatory to excitatory pathways in our circuit reconstruction [Markram et al., 2015]. The protocols have been chosen to be representative of typical low-, medium- and high-frequency stimulation in standard STDP experiments [Markram et al., 1997b, Sjöström et al., 2001].

The STDP maps for the three protocols are shown in Fig. 3.19. As expected only few pathways showed any change after the 2 Hz protocol, while most of them potentiated at 50 Hz stimulation. The 10 Hz protocol produced mixed outcomes, with a tendency for depression. Furthermore, we observed some pathways which did not potentiate under any circumstance, i.e. layer 2/3 PC to layer 4 SP connection.

The variability across pathways is even more evident when comparing the classical frequency-dependence curves (see Fig. 3.20). As previously mentioned, the 10 Hz protocol is particularly interesting because it differentially induces LTP, LTD or no change depending on the identity of the pre- and post-synaptic neurons.

[6] We do not consider studies lacking a precise identification of the pre- and post-synaptic cells, i.e. studies employing extracellular electrodes for axonal stimulation.

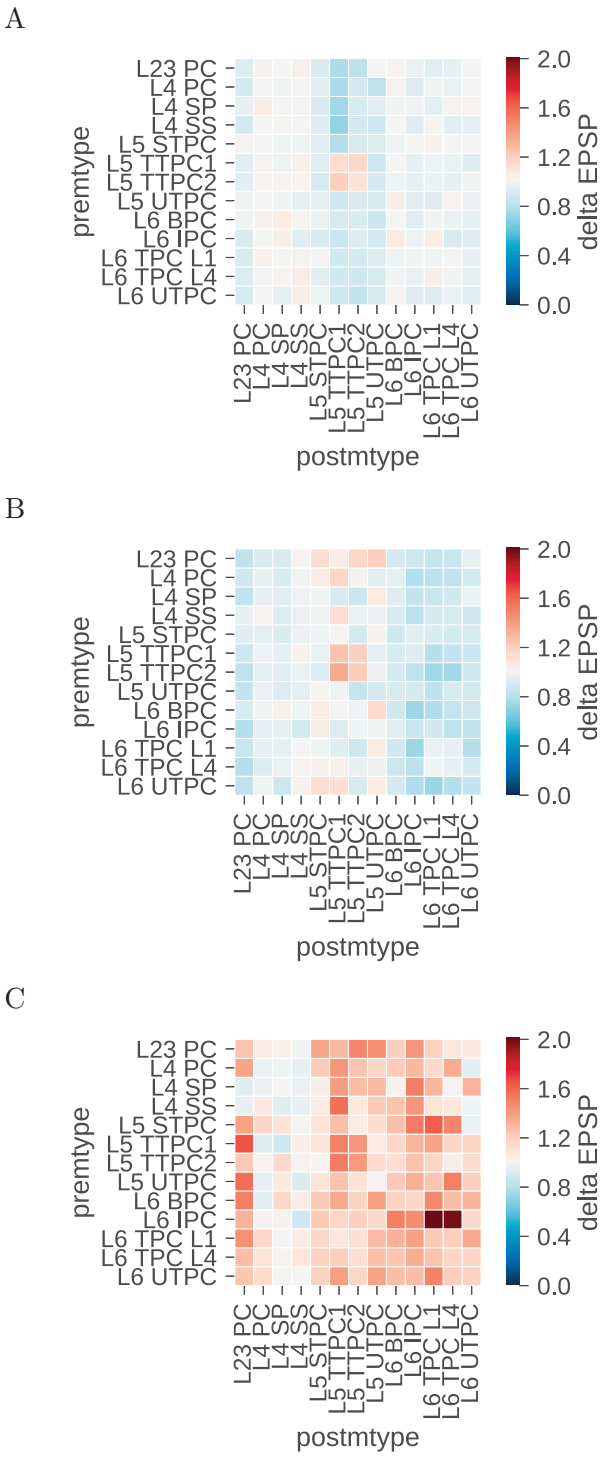


Figure 3.19 – STDP map. A Stimulation at 2 Hz. B Stimulation at 10 Hz. C Stimulation at 50 Hz.

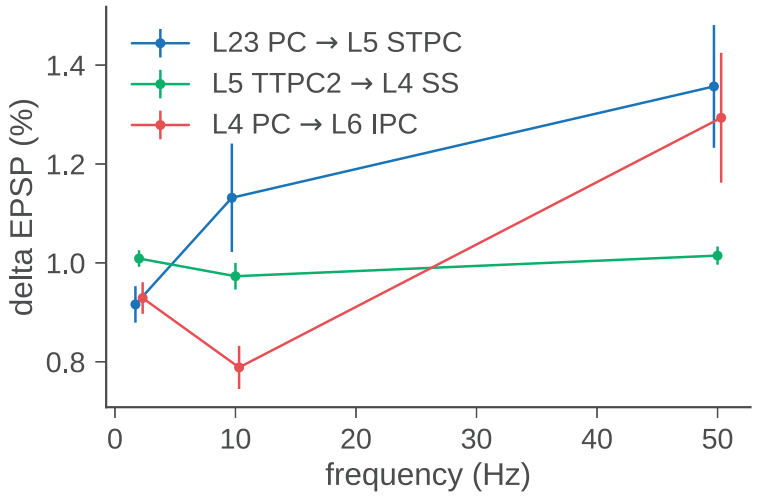


Figure 3.20 – LTP frequency dependence for uncharacterized pathways.

3.4 NMODL implementation and HPC considerations

As mentioned at the beginning of this chapter, GluSynapse was implemented in NEURON model description language (NMODL) for the NEURON simulator [Carnevale and Hines, 2006]. Here we discuss some of the technical details of model implementation and simulation. In particular, in Sec. 3.4.1 we describe how to account for dendritic spines without affecting simulation performance. In Sec. 3.4.2 we review integration methods in NEURON and our strategy to reduce simulation time. Finally in Sec. 3.4.3 we propose an efficient implementation of the synapse elimination mechanism.

This section often relies on some basic knowledge of the NEURON simulator's internals. For this reason, we encourage the reader to consult *"The NEURON Book"* [Carnevale and Hines, 2006] whenever necessary.

3.4.1 Spine as virtual compartment

The most natural way to model dendritic spines in NEURON is to use one or more dedicated compartments. A popular approach consists of one compartment for the spine neck and one for the head. This choice allow to control in great details several aspects of spines' internal dynamics, such as diffusion or neck resistance. However, in large network simulations it is practically unfeasible to explicitly model every spine. A typical pyramidal neuron consists of a few hundreds of compartments but features several thousands of spines. NEURON heavily rely on the assumption that the matrix of compartments is relatively small and easy to invert. The explosion of the number of compartment increases the complexity of the problem quite dramatically.

For this reason, we did not implement spines as actual compartments bur rather as *virtual* ones. That is, the synapse model itself takes care of every aspect related to the spine, without the need to add more compartments to the simulation: Ionic diffusion is neglected and calcium dynamics are explicitly managed (see 3.1.5); VDCC are implemented as a point mechanism and current density estimation is carried on internally, as opposed to the usual density mechanisms inserted into the membrane and directly managed by NEURON; calcium compartmentalization do not rely on NEURON's internal features for ionic currents to avoid affecting the parent dendrite; Spine neck is neglected as it seems to have a small impact in our simulations (data not shown).

Due to the extended "duties" of the synapse model, Glusynapse can be considered more of a synaptic simulator within NEURON, rather than just a regular model. Overall this approach allows our simulations to be as efficient as those without spines and to capture most of the spine's aspects relevant for synaptic plasticity.

3.4.2 GluSynapse integration methods

Before digging into the integration methods used in this work, a premise on what is available in NEURON seems required. For an exhaustive review of NEURON's integration methods and a deeper analysis of numerical error, please refer to *"The NEURON Book"* [Carnevale and Hines, 2006].

NEURON offers several fixed time step integrators, specified in NMODL at model design time:

cnexp An implicit method for solving linear, non coupled, ODE of type $\dot{x} = f(v, x)$. This method is particularly efficient and relies on the existence of an analytical solution, i.e. $x = e^{-\frac{t}{\tau}} + c$;

derivimplicit An implicit method for solving general ODE. This method trades performance for robustness and range of applicability;

euler An explicit method based on forward Euler. This method is extremely fast but its usage is discouraged, as it requires very small time steps to prevent numerical instability and errors accumulate over time.

Other choices are available, but they are either less common (i.e. **runge** method) or designed for special purposes, such as solving kinetic schemes (i.e. **sparse** method).

NEURON provides also two adaptive integrators: CVODE by SUNDIALS [Hindmarsh et al., 2005] and DASPK [Brown et al., 1994]. The former is particularly interesting because it supports local variable time step, a feature that can dramatically improve the performance in certain contexts, such as simulations involving few AP to propagate[7].

Network simulations performed at the BBP involve a large number of spike exchanges and for this reason they are not suited for adaptive integration, at least for now. Standard, non-plastic, synapses are then integrated using the fast **cnexp** method, as their equations satisfy the constraints imposed by this solver. GluSynapse instead is a much more complex model, involving non-linear and/or coupled differential equations, and requires the **derivimplicit** integration method. Unfortunately, the introduction of GluSyanapse in the network model caused a major performance drop which could not be tolerated, as it increased the computational requirements beyond our current resources. To solve the problem, we decided to temporarily switch to the **euler** integrator for GluSynapse. We assumed that the risk of numerical instability would be mitigated by the small integration step used in all our network simulations ($dt = 0.025$ ms) . Furthermore, we explicitly verified that the integration error is small compared to the standard precision of our simulations (data not shown).

The **euler** integrator was only used for network simulations. Model fitting (Sec. 3.2) instead was a perfect use case for the fast variable time step implicit integrator CVODE.

3.4.3 Inhomogeneous Poisson elimination

Synapse elimination was modeled as an inhomogeneous Poisson process, as the pruning probability is a function of synaptic efficacy (see Sec. 3.1.7). For this reason, we cannot assume an exponential distribution for the event intervals. How to calculate the intervals for an inhomogeneous Poisson process is an active area of research and we do not have

[7] Spike exchange is a global synchronization barrier: after spike delivery at the postsynaptic side, the system needs to be re-initialized and every "jump" performed by the adaptive solver to be undone. For this reason, network simulations might not benefit from local variable time step integration if too many spikes are generated.

a general formula yet. However, under certain assumptions, we can approximate the inter-arrival interval as an exponential distribution.

The first obvious approach would be to draw a random sample every time step, as the synaptic efficacy variable ρ evolves on much longer time scales and it could be considered constant in between sampling. Unfortunately drawing so many random numbers would be too expensive to be practical.

A better solution can be designed if we consider that (a) the average firing frequency in vivo is approximatively 1 Hz and (b) abrupt changes in ρ require the NMDAR to be involved. That is, in between spikes the ρ variable can be assumed to be constant, as the inter-spike interval is approximatively 1 s and any extreme change in ρ happens almost instantaneously after an incoming spike. This approach greatly reduces the number of random sample to be drawn without introducing a big modeling error.

4 Effects of functional synaptic plasticity on neural circuits

"Too much light often blinds gentlemen of this sort. They cannot see the forest for the trees."

Christoph Martin Wieland,
Musarion

In this chapter we describe the impact of introducing functional synaptic plasticity in a digital reconstruction of neocortical tissue [Markram et al., 2015]. We simulated the same neural network under two conditions: spontaneous activity and thalamocortical stimulation. In the former case, the circuit was left free from external stimuli for the whole duration of the simulation (1050 s). In the latter instead, we provided a external stimulus through the innervating thalamocortical fibers. The circuit was simulated at in vivo-like calcium levels (1.2 mM).

In the next sections we describe how the circuit and the stimuli were defined (Sec. 4.1). Then we present several preliminary analyses, required to gather initial information on network behavior during plastic simulations (Sec 4.2). Finally, we show how the introduction of functional plasticity induced strengthening of connections between neurons similarly innervated, as previously observed in vitro [Perin et al., 2011] (Sec. 4.3).

4.1 Fundamentals of microcircuit reconstruction and simulation

The process of reconstructing and simulating a neocortical microcircuit is described elsewhere Markram et al. [2015]. In brief, reconstructed neurons were placed in space according to data on dendrites, proportions and morphological features. Electrical types were assigned by fitting the reconstructed neurons to in vitro electrophysiological recordings [Druckmann et al., 2007, Hay et al., 2011, Van Geit et al., 2016]. Neurons were connected using a biophysically constrained probabilistic algorithm [Reimann et al., 2015]. Synaptic physiology was assigned to match the known features of each viable synaptic pathway.

The microcircuit used in this work is composed of 224000 neurons, organized in seven cortical columns and six layers (Fig. 4.1A). The columns are arranged in a center-surround fashion: a single central column, the main subject of all our analyses, is surrounded by six other columns, providing a realistic neural environment without major

edge effects. All the cortico-cortical excitatory synapses in the central column are plastic, while those in the other columns, the thalamocortical synapses and all inhibitory synapses are not. Thalamocortical fibers innervate the seven columns matching known constraints on target layers, synaptic physiology and number of synapses per connection [Markram et al., 2015]. To provide spatially defined stimuli to the network, we bundled the fibers in 100 clusters, based on the fibers position (Fig. 4.1). Fibers belonging to the same cluster are assumed to carry the same information into the cortex (see Sec. 4.1.1) The microcircuit was simulated using the NEURON simulator on the MIRA and

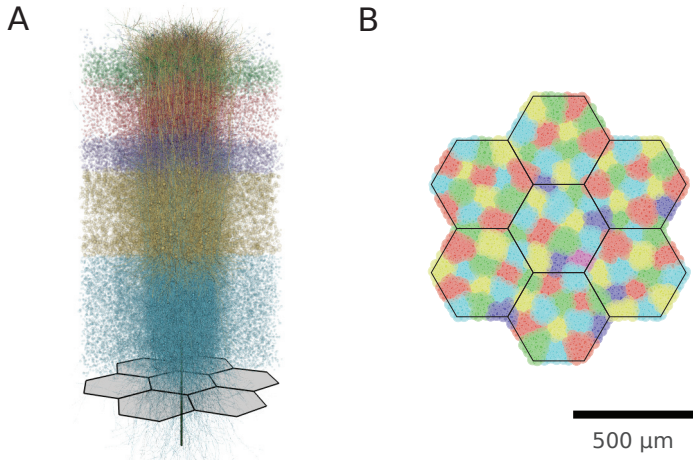


Figure 4.1 – Circuit layout. A, the circuit used in our simulations is composed by seven columns organizes in a mosaic. Excitatory synapses in the central column are plastic. Thalamocortical fibers (one exemplar showed as a green line) innervates the circuit matching several biological constraints [Markram et al., 2015]. B, neighbor thalamocortical fibers are functionally clustered into 100 groups. Fibers belonging to the same cluster are always stimulated together and represent a "pixel" of the network global receptive field.

BBPIV supercomputers.

4.1.1 Thalamocortical stimulation

Thalamocortical stimuli were generated using a custom Python module. Each stimulus is fed to the generator as a dotted pattern impinged on the microcircuit mosaic (see Fig. 4.2). The stimulus generator produces Poissonian spike trains for every fiber. The firing rate of every fiber is determined by the presence or absence of a "dot" in its receptive field.

The whole set of stimuli used in this work is shown in Fig. 4.3. We generated eight different stimuli with identical statistical properties (i.e. number of dots, firing frequency). During simulation, the eight patterns were presented in a random order every 200 ms. Every 30 s we left the network unstimulated for 1 s to test the effects of synaptic depression on the stimulus. That is, thalamocortical fibers could recover from

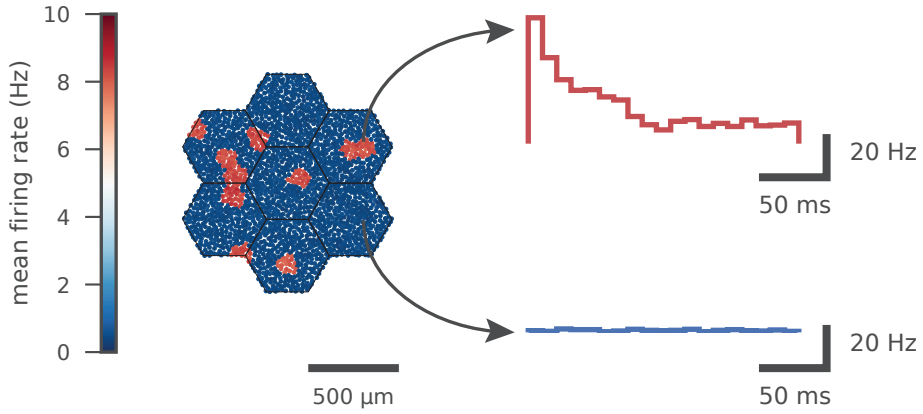


Figure 4.2 – Exemplar stimulus. A randomly generated dotted pattern is converted into a Poissonian spike train for every thalamocortical fiber. Fibers having a dot in their receptive fields rapidly increase their firing rate in the first 75 ms after the presentation of the pattern (red firing rate histogram). Unstimulated fibers have instead a low and constant firing rate (blue firing rate histogram).

depression and provide a stronger stimulus to the cortex, emulating the abrupt arrival of a stimulus.

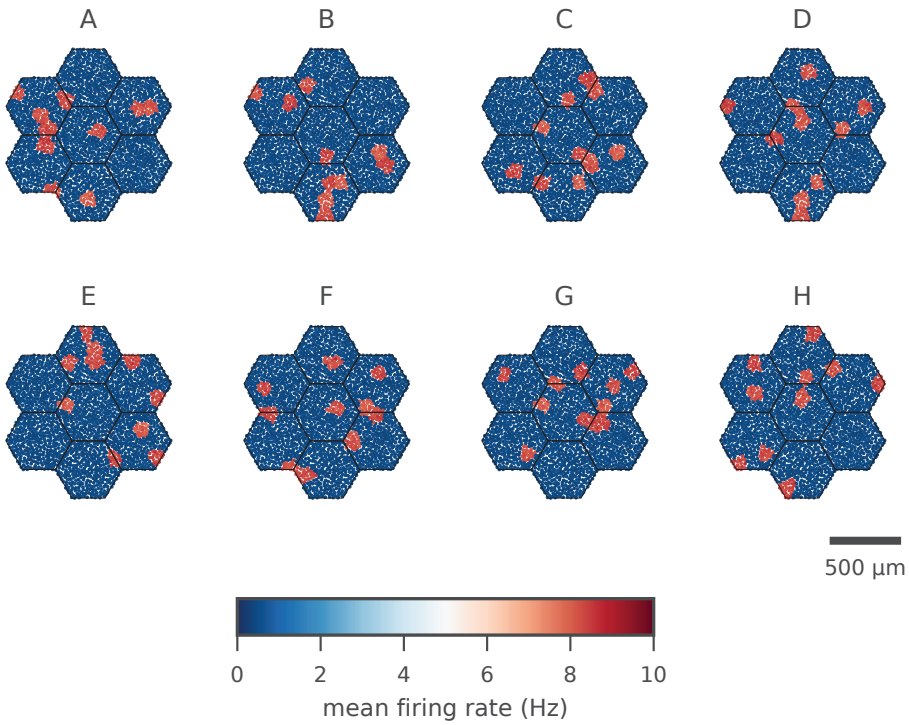


Figure 4.3 – Summary of all stimuli. Eight stimuli were randomly generated and labeled alphabetically. The stimuli share the same statistics in terms of intensity and stimulated area.

4.2 Preliminary analyses

Before evaluating the computational implications of having functional synaptic plasticity in the network, we carried on several preliminary analyses. Our goal was to establish whether the direction and magnitude of plastic changes have any global trends in the network, to identify regions of interest and possible boundary conditions to take into account.

4.2.1 Network balance

The main concern in any network simulation featuring Hebbian-like excitatory plasticity is the tendency to runaway excitation, due to the disruption of excitation-inhibition balance. To prevent this effect, these models are usually coupled with some sort of weight normalization mechanisms, resembling an homeostatic process. Although what we presented in this work is not a traditional Hebbian model, we cannot rule out a priori the possibility of runaway excitation. To verify the stability of excitation-inhibition balance in out simulations, we analyzed the evolution of firing rate over time in both the spontaneous and stimulated condition.

In the spontaneous case, activity levels remained constant throughout the whole du-

ration of the simulation (Fig. 4.4). The overall firing rate almost never exceeded 2 Hz, except for the initial network burst. This event was caused by the fully recovered state of all the synapses at the beginning of every simulation. We did not try to prevent network bursts as they are, in limited amount, part of the normal functioning of a neural network. Runaway excitation rather refers to the generation of these events at high frequency, preventing the circuit from doing anything else.

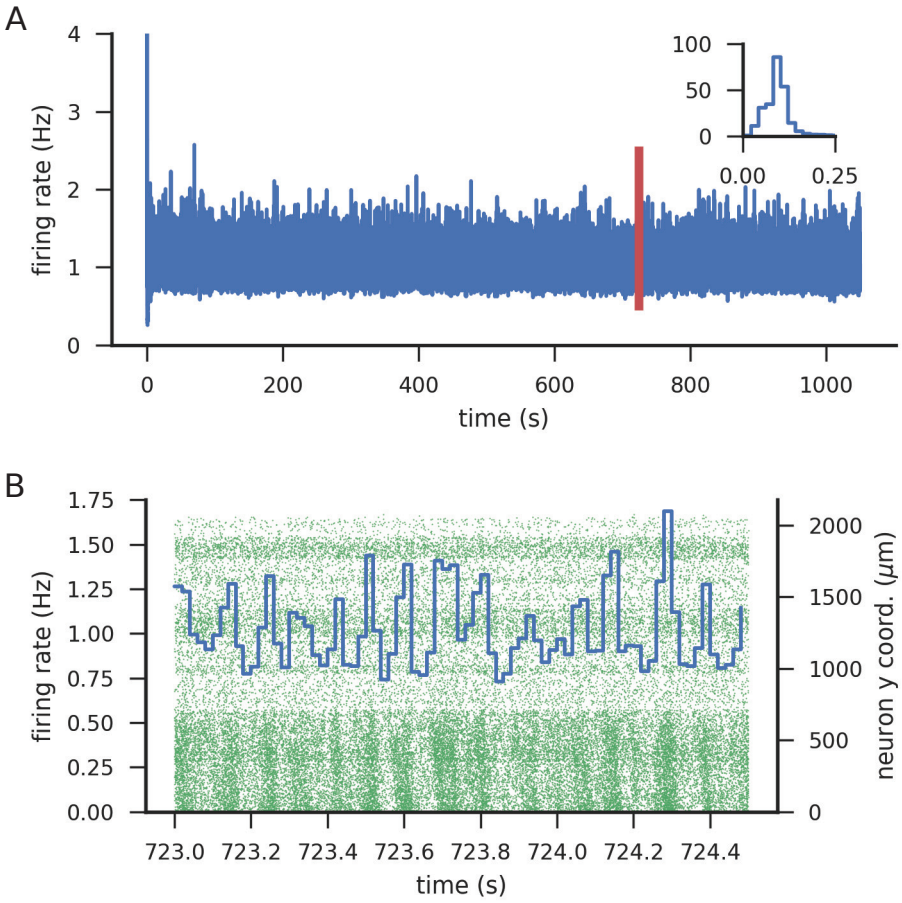


Figure 4.4 – Firing rate evolution during spontaneous network activity. A, Firing rate over time for all cells in the central column (bins size 20 ms); the y axis was truncated to 4 Hz to better resolve the low firing rate; red square marks the time window in B; inset shows the initial network burst, axis units as in the main panel. B, close-up on network activity in the red window marked in A; blue line shows the firing rate histogram (left y axis, truncated at 1.75 Hz); green dots show the spikes of individual neurons, sorted by depth in the tissue (right y axis).

During thalamocortical stimulation, the circuit showed a more marked tendency to burst (Fig. 4.5). However, the periodicity of these events did not reach alarming levels

(see network burst in Markram et al. [2015]). Vice versa, to our surprise, they acted as an intrinsic homeostatic mechanism. As we later show in Sec. 4.2.2, network busts bring a large fraction of layer 5 Thick-tufted pyramidal cells (TTPC) synapses down to an intermediate efficacy range. This effect seems to prevent uncontrolled potentiation and so the explosion of network activity. We then tested weather a specific stimulus

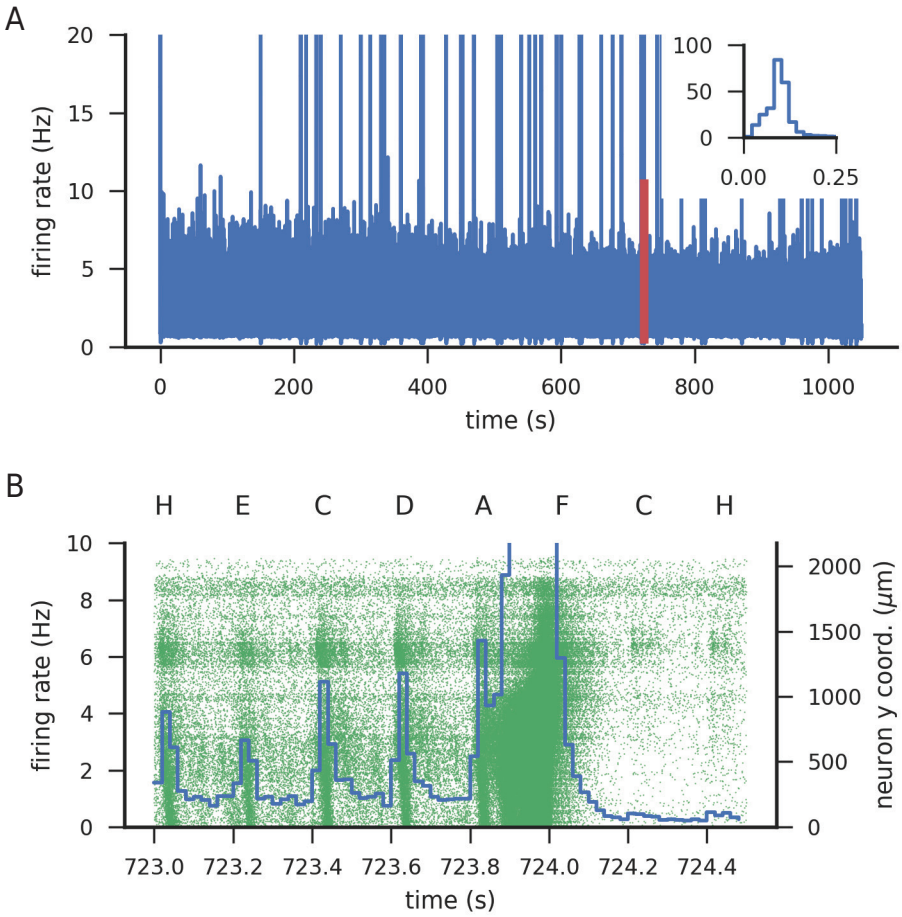


Figure 4.5 – Firing rate evolution during stimulated network activity. A, Firing rate over time for all cells in the central column (bins size 20 ms); the y axis was truncated to 20 Hz to better resolve the low firing rate; red square marks the time window in B; inset shows the initial network burst, axis units as in the main panel. B, close-up on network activity in the red window marked in A; blue line shows the firing rate histogram (left y axis, truncated at 10 Hz); green dots show the spikes of individual neurons, sorted by depth in the tissue (right y axis); letters on top mark the stimulus applied at the corresponding time.

was more likely to generate a burst. Indeed half of them were generated by the stimulus labeled as *A* (Fig. 4.6). We are currently investigating why this specific input pattern was burst-prone, but no obvious reason was found yet.

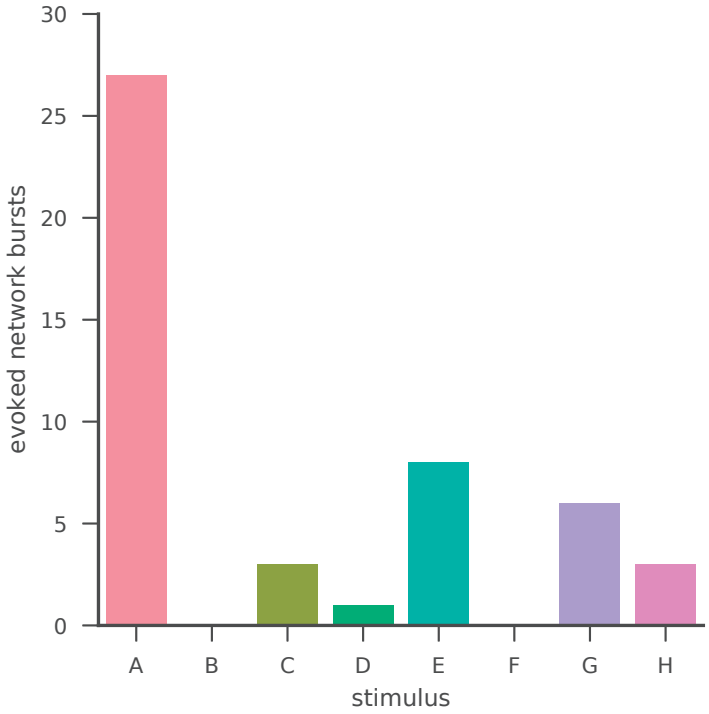


Figure 4.6 – Burst incidence per stimulus. Stimulus pattern "A" triggered half of the total network bursts.

The careful reader will have noticed that our model of glutamatergic synapse lack an explicit homeostatic component. We believe that the need of an homeostatic term to balance most Spike Timing Dependent Plasticity (STDP) learning rules has created the false expectation of a specific biological mechanism to counterbalance the emergence of strong weights and runaway excitation. Based on our data, we suggest that the picture is actually simpler than that. The network itself tend to stabilize its state, as network bursts effectively dump the "weights". Obviously we do not question the existence of homeostatic plasticity as a mechanism to rescue from catastrophic events, such as brain lesions. We instead question the need of an homeostatic term to explain normal network functioning. One might argue that our model does indeed have an homeostatic mechanism, as the release probability cannot grow above the prescribed bounds (see Sec. 3.1.6). However, we believe that assuming the existence of physiological bounds for synaptic properties, such as release probability, is rather different than the common weight normalization.

4.2.2 Redistribution of synaptic efficacy

We then examined how single synapses on the same postsynaptic neuron evolved over time. For this analysis we considered a typical layer 5 TTPC. In the spontaneous case the

vast majority of synapses converged to their fixed points within the time of simulation (Fig. 4.7). For this reason the initially unimodal distribution of release probabilities became strongly bimodal. During thalamocortical stimulation, synapses showed more

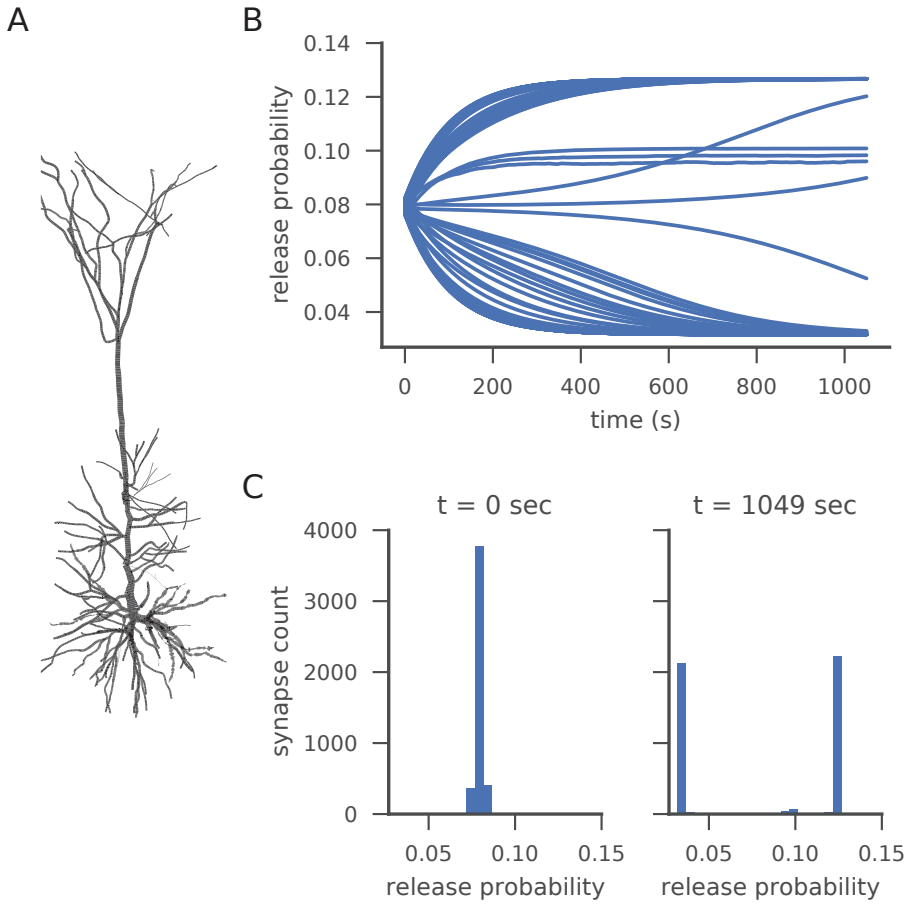


Figure 4.7 – Redistribution of synaptic efficacy during spontaneous activity. A, 2D morphological reconstruction of the cell used in this analysis. B, release probability evolution for a small sample of excitatory synapses on the cell (3 % shown). C, release probability distribution before (left) and after (right) spontaneous activity simulation.

interesting dynamics (Fig. 4.8). In particular, a third (apparent) equilibrium point appeared for the release probability variable. This new point, approximatively located at 75 % of the release probability range, attracted a large fraction of the available synapses. It was interesting to notice that the existence of this point became clear after the second network burst (see 4.5). To complement our observations, we analyzed the spatial distribution of synapses on the cell morphology after simulation (Fig. 4.9). We found that synapses having release probability within a standard deviation from the mean were mostly located on the basal and oblique dendrites. This finding was

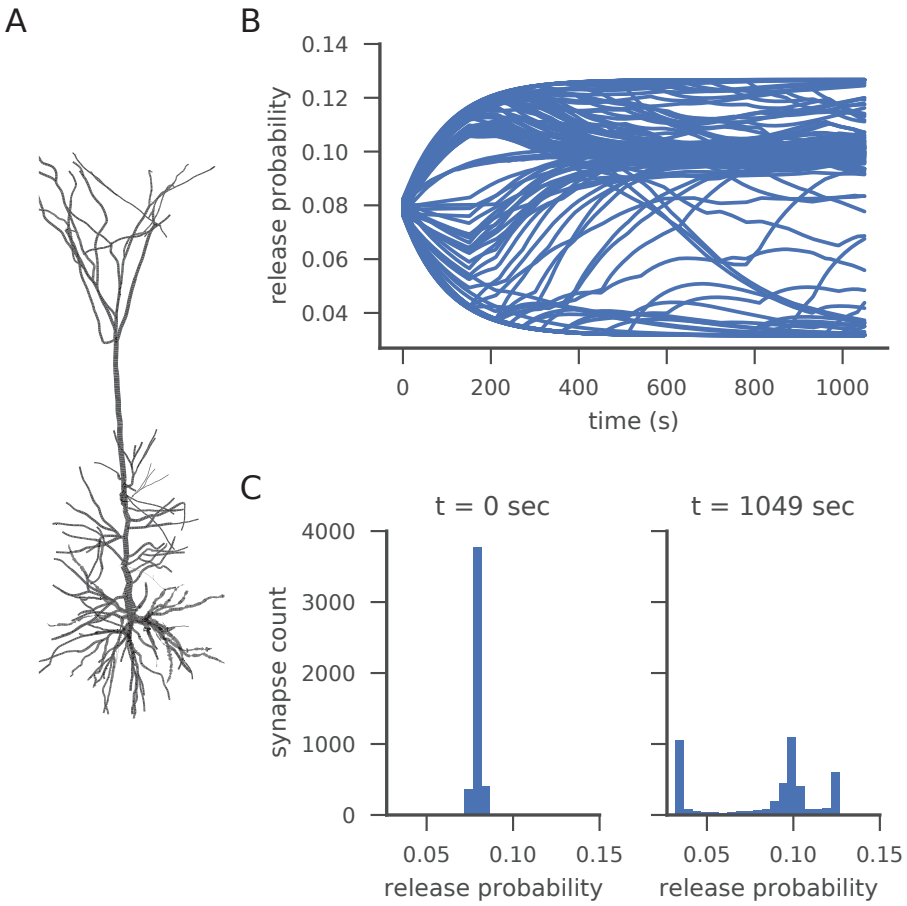


Figure 4.8 – Redistribution of synaptic efficacy during thalamocortical stimulation. A, 2D morphological reconstruction of the cell used in this analysis. B, release probability evolution for a small sample of excitatory synapses on the cell (3 % shown). C, release probability distribution before (left) and after (right) spontaneous activity simulation.

particularly interesting because it follows the characteristic innervation pattern of layer 5 TTPC to layer 5 TTPC [Markram et al., 1997a]. We are currently investigating what functional implications this peculiar distribution might have.

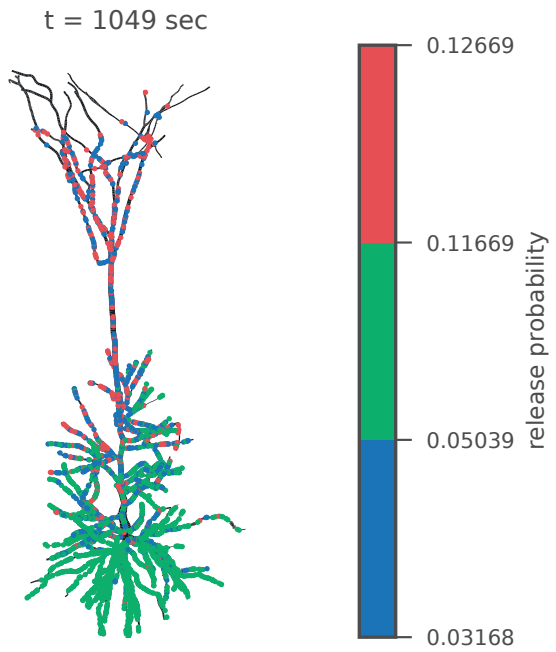


Figure 4.9 – Synaptic efficacy distribution of a typical layer 5 TTPC. Synaptic locations are marked by circles and divided in three groups, depending on release probability. Weak synapses in blue (below one standard deviation from the mean); average synapses in green (within one standard deviation from the mean); strong synapses in red (above one standard deviation from the mean). Most average release probability synapses are located on the basal and oblique dendrites.

4.2.3 Layer specific changes

We next examined how the origin layer of each cell affects the average plastic change (see Fig. 4.10). In the spontaneous case, the mean release probability did not change in any layer, while the variance increased for all of them. This is a clear sign that these synapses are simply converging to their stable fixed points without major reorganization (see Sec. 4.2.2). In the stimulated case instead, only layer 2 and layer 3 remained "neutral". The result was largely expected, as firing activity in layer 2 and 3 is very low even during stimulation and so unlikely to cause large plastic changes. This phenomenon is partially due to physiological features of neurons in these layers and partially to the missing lateral spread of axons, not yet fully captured in our network model. The lower layers are instead much more engaged, as reflected by the smaller variance change. Overall, layer 5 cells showed a tendency to potentiate while layer 4 and 6 cells to depress. For an in depth analysis of pathway-specific changes see Sec. 4.2.2.

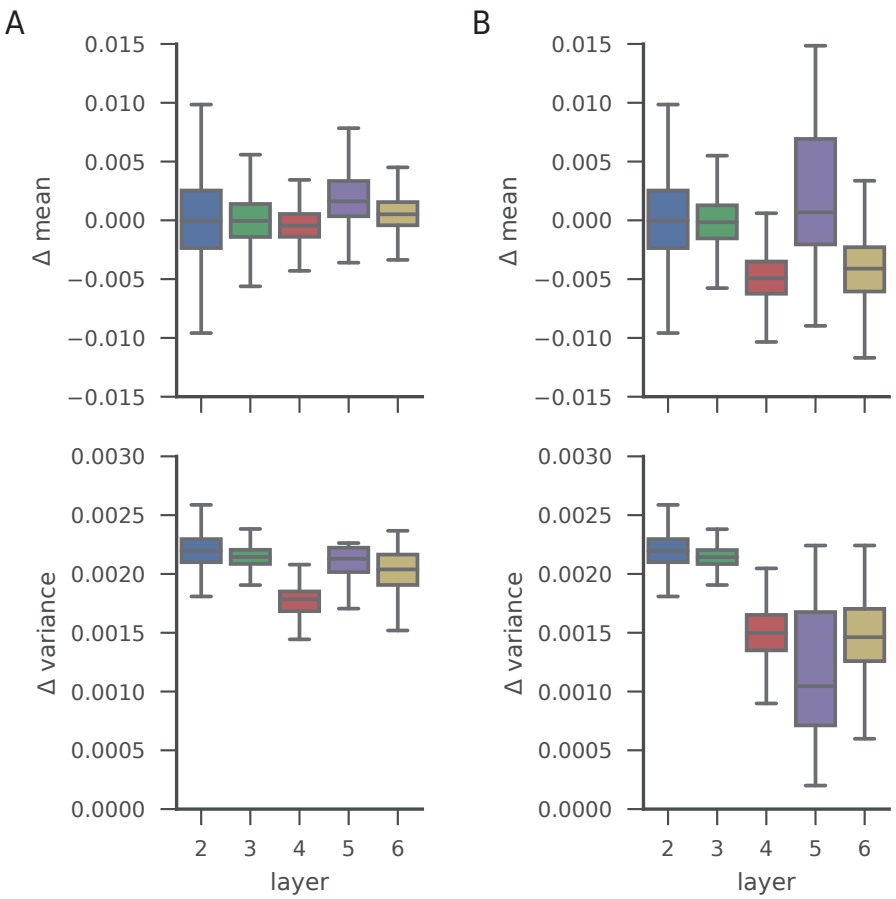


Figure 4.10 – Average release probability change after 10 minutes of simulation. A spontaneous activity simulation. Mean (top) and variance (bottom) of the neuron average change, grouped by soma layer. B same of A but for the stimulated case.

4.3 Common neighbors strength bias

It has been shown that the Excitatory Post-Synaptic Potential (EPSP) amplitude between connected pyramidal cells in layer 5 is correlated with the number of *common neighbors* Perin et al. [2011]. Given any pair of cells i and j, a common neighbor is a cell k connected to both i and j (see Fig. 4.11A). That is, the average connection strength increases with the number of common neighbors. Whether the common neighbors strength bias is genetically prescribed or the consequence of plastic mechanisms at work is currently unknown.

The digital reconstruction of a neocortical circuit used in this work satisfies several constraints on synaptic physiology (see Markram et al. [2015]), but the common neighbors strength bias is not one of those. To test the hypothesis that this bias arises

spontaneously in a plastic neural network, we analyzed the relationship between number of common afferent[1]and mean release probability for a subset of TTPC connections in layer 5. A pool of 300 connections was randomly selected from the central column. Common afferents instead were estimated considering all layer 5 TTPC, without any restriction on the cortical column. To assess whether number of common afferents and mean release probability were correlated, we considered the Spearman's rank correlation coefficient [Zwillinger and Kokoska, 2000]. This measure was chosen as it does not require any assumption on the marginal distributions of the paired variables to test (i.e. normality). We found that in both the stimulated case (Fig. 4.11) and spontaneous activity (Fig. 4.12) the bias emerged at the end of the simulation (t = 1050 s), particularly strong in the former case.

[1] A common afferent is simply a common neighbor satisfying the property of being pre-synaptic to both the target neurons. Perin et al. [2011] showed that the strength bias is even more marked when common afferents rather than common neighbors are considered.

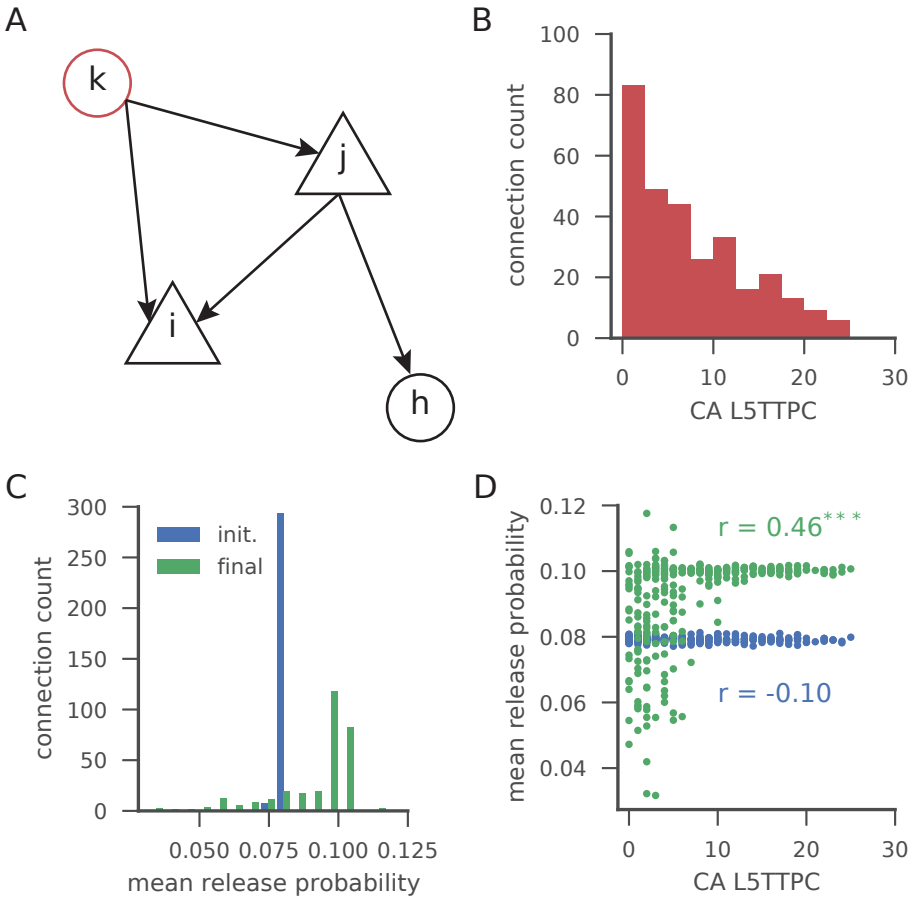


Figure 4.11 – Emergence of common neighbors strength bias (talamocortical stimulation). A, example of common neighbor; neuron k is both a common neighbor (CN) common afferent (CA) for neurons i and j, while neuron h is not. B, common afferent (CA) distribution for the 300 connections considered in this analysis; CA were computed considering the full population of layer 5 TTPC. C, mean release probability distribution before (t=0) and after (t=1050) simulation. D, common neighbors strength bias; after simulation a strong correlation between common afferents and mean release probability emerged.

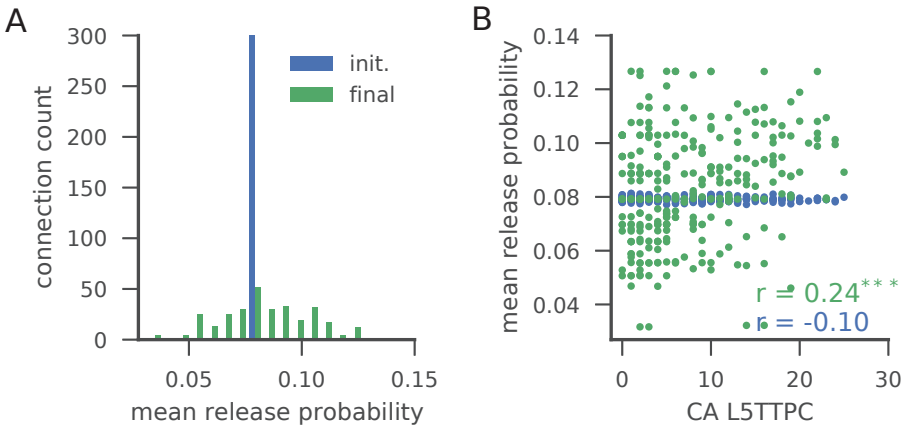


Figure 4.12 – Emergence of common neighbors strength bias (spontaneous activity). A, mean release probability distribution before (t=0) and after (t=1050) simulation. B, common neighbors strength bias; after simulation a weak correlation between common afferents and mean release probability emerged.

5 Conclusion

"If you can meet with Triumph and Disaster and treat those two impostors just the same."

RUDYARD KIPLING, *If*

In this work we integrated state of the art data and theories on excitatory synaptic plasticity to design a comprehensive model, suitable for large scale simulations of neocortical circuits. Our main goal was to unify the fragmented knowledge on the subject and to produce a centralized resource to study the biological foundation of learning and memory in a computer-based framework. We started by sketching a simplified description of the synapse life-cycle on the basis of available experimental literature (see Chapter 2). The model consists of a discrete set of states and transitions. Each state represent a well defined axo-dendritic configuration, such as *depressed synapse* or *bouton-filopodium apposition*. Transitions between states model the mechanisms of synaptic plasticity, the main subject of this work. From the high-level description of the synapse life-cycle we derived a mathematical model of plastic synapse and implemented it in NEURON (see Chapter 3). Whenever possible, we relied on existing components and theories from literature, limiting extensions to the minimum required. The Long Term Potentiation (LTP)/Long Term Depression (LTD) component of the model was then parameterized using a custom Genetic Algorithm (GA)-based strategy to reproduce in silico the effects of three standard STDP protocols on layer 5 TTPC connections [Markram et al., 1997b]. The parameterization obtained for layer 5 TTPC connections was generalized to all other excitatory-to-excitatory pathways. Small variations of the fitted parameters were necessary to adapt the model to the synaptic physiology of the other pathways. Finally, we tested the effects of functional and structural synaptic plasticity on neural circuits by simulating a large-scale reconstruction of neocortical tissue [Markram et al., 2015]. We considered two network conditions: spontaneous activity and thalamocortical stimulation (see Chapter 4).

The integrative nature of this thesis led us to challenge and revise many common assumptions on synaptic plasticity. For example, accounting for stochastic vesicle release and postsynaptic calcium dynamics had profound implications on the requirements for LTP induction. The Graupner-Brunel (GB) model [Graupner and Brunel, 2012], revisited in this work, explains the LTP's frequency-dependence by postulating postsynaptic calcium accumulation at high frequency-stimulation. This idea is an extension of the popular "calcium control hypothesis" [Lisman, 1989]: prolonged but moderate elevation of intracellular calcium triggers LTD, while larger transients result in LTP. However, postsynaptic calcium accumulation is incompatible with the physiological features of

neocortical synapses. That is, fast calcium buffering dynamics, combined with low presynaptic release probability and a small Readily-Releasable Pool (RRP), result in rapid calcium clearance at the spine, even at high-frequency stimulation (see Sec. 3.1.6 for more details). We then concluded that the interplay of several calcium buffers is a requirement for LTP's frequency-dependence, in line with the results of other studies [Holmes and Levy, 1990, Markram et al., 1998] (for a review see [Manninen et al., 2010]). Our model captured LTP's and LTD's frequency dependence reported by Markram et al. [1997b], Sjöström et al. [2001], without explicit fitting on these properties. Furthermore, it reproduced the results on distance-dependent synaptic plasticity [Sjöström and Häusser, 2006], even though this dataset was also never used during fitting. In network simulation, we observed the strengthening of connections between neurons similarly innervated, as previously shown in vitro [Perin et al., 2011]. These results show how our approach, based on the systematic enforcement of biological constraints, actually simplifies the problem of modeling synaptic dynamics: a small input dataset and a few new constraints were sufficient to capture a broad range of long-term plastic dynamics for several pathways.

The main product of this work is a simulation-based framework to study synaptic plasticity in morphologically detailed neural circuits. We hope that the availability of such a resource will trigger more theoretical and experimental work on the subject. In particular, our framework could be used to test specific hypotheses on network functioning during stimulation, other than the one we are already considering. Furthermore, the modular structure of the synapse model allows the replacement of any component with a new formulation. For this reason, GluSynapse could serve as a scaffolding environment for the design of other models of specific synaptic mechanisms. Finally, experimental work could be simplified by testing in the network model the expected outcome of a large set of protocols. This approach could greatly reduce the burden of experiment design and guide the intuition of researchers even during the experiments.

Regardless of the specific implementation, choices and assumptions discussed in this thesis, we believe that the underlying idea of this work has an important scientific value. The brain is an organ and a dynamical system of exquisite complexity and, as such, virtually any idea and theory about its functioning would find some experimental support. For this reason, the careful integration of data and theories from different sources is a crucial ingredient for well founded hypotheses on brain functioning. We attempted to reduce the complexity of investigating synaptic plasticity by looking for mismatches and limit cases. As previously mentioned, this approach forced us to re-evaluate the impact of many well accepted assumptions on synaptic plasticity and, eventually, to reject some of those. We have no doubt that also this model, like every other, is wrong at a certain level of detail. However, we think it represents a significant step towards a better understanding of synaptic plasticity and its implications on neural circuits.

5.1 Limitations

In this work we considered plasticity of excitatory-to-excitatory connections only. We decided to focus on these connections because they are the most abundant and best characterized, both theoretically and experimentally. However, plasticity of the other three "families", namely excitatory-to-inhibitory, inhibitory-to-excitatory and inhibitory-to-inhibitory, certainly have a role in higher order brain functions such as learning and memory.

Although the excitatory-to-inhibitory case seems to be a minor extension, the other two types have their own bulk of literature, data and theories and so represent an entirely new subject. The structural plasticity component of the synapse model is still a work in progress. For the mathematical description, we relied on the work of Fauth et al. [2015] with minor modifications. This formulation was particularly convenient because it guarantees the emergence of bimodal distributions for the number of synapses per connection, under certain assumptions on the functional plasticity model. Assumptions all met by the GB model, also used in this work. However, despite the mathematical proof provided by Fauth et al. [2015], we would need to test this hypothesis in simulation, as our model of functional changes departed quite substantially from the original formulation by Graupner and Brunel [2012]. Furthermore, to improve the current model parametrization we would need a larger experimental dataset, focusing on the specificity of synaptogenesis and synapse elimination. That is, we do not have quantitative data linking the emergence (disappearance) of a particular synapse with the activity of its pre- and postsynaptic neurons.

5.2 Future work

The first release of the GluSynapse model was centered around data integration. We focused on capturing the most fundamental synaptic properties and orchestrating the different mechanisms in a coherent manner. In the second iteration will instead concentrate on parameter refinement.

The N-methyl-D-aspartate receptor (NMDAR) plays a crucial role in the induction of functional synaptic plasticity. For this reason, accurately modeling its dynamics is of primary importance. In this work we relied on the formulation proposed by Jahr and Stevens [1990]. Despite popularity, this model has some important drawbacks. First, the model is parametrized on hippocampal culture data, without any compensation for liquid junction potential. Second, the model does not properly capture NMDA spike generation. We plan to address these issues in the next iteration of our GluSynapse, based on a set of recent studies on NMDAR dynamics. The model refinement will be carried on in two phases. Initially we will simply re-parametrize the current model using neocortical data from in vitro experiments [Vargas-Caballero and Robinson, 2003]. In the final stage of the project, we will consider more detailed descriptions of the NMDAR [Vargas-Caballero and Robinson, 2004, Moradi et al., 2013].

The expression mechanisms of functional plasticity in the somatosensory cortex are

still debated. In particular, it is not completely clear whether STDP stimulation triggers pre- or post-synaptic changes. We assumed the former case, based on a set of LTP experiments employing an induction protocol similar to STDP [Markram and Tsodyks, 1996]. However, more recent experiments in the visual cortex seem to indicate both a pre- and post-synaptic locus of expression [Sjöström et al., 2007]. To resolve this mismatch, we are currently conducing an in depth study of STDP expression mechanisms in the somatosensory cortex [Perin and Chindemi, in preparation]. The results of our in vitro experiments will have important implications on the next refinement of GluSynapse. In particular, preliminary analysis of our data suggests the presence of a postsynaptic component, complementing the (expected) presynaptic one. Adding a postsynaptic term to our synapse model is then a necessary extension. The main effect of this change will be a reduction of the distance between the depressed and potentiated fixed points for the release probability, $U_{\mathrm{SE}}^{(d)}$ and $U_{\mathrm{SE}}^{(p)}$ respectively. That is, assuming a purely presynaptic expression mechanism forced us to allow a large difference between $U_{\mathrm{SE}}^{(d)}$ and $U_{\mathrm{SE}}^{(p)}$ to capture the full range of EPSP changes in our dataset. The introduction of a postsynaptic component will obviously relax this constraint. Furthermore, the new data currently acquired in our lab will provide a quantitative estimate of the relative contribution of the pre- and post-synaptic contribution.

The first release of GluSynapse featured fixed values for $U_{\mathrm{SE}}^{(d)}$, $U_{\mathrm{SE}}^{(p)}$ and N_{RRP}. We plan to transition from these mean values to more biologically plausible probability distributions. This refinement is required to better capture several aspects of synaptic reliability [Rahmon et al., in preparation]. Work on N_{RRP} is already at a mature stage and we only miss an analysis of the relationships with other synaptic parameters. That is, we are currently investigating whether N_{RRP} is independent from the other synaptic variables (i.e. single vesicle release probability or postsynaptic area) or a tight coupling exists. Furthermore, we need to establish its role in the expression of long-term functional changes. The second release of GluSynapse will be parametrized using a new generalization algorithm (see Sec. 3.2.5). The existing method does not allow to impose soft constrains, such us penalizing very large or small parameters, and it is very sensitive to initial conditions. The new algorithm is being designed to overcome these issues and to accommodate all the changes described above.

In this work we did not include any network simulation featuring rewiring. Simulating rewiring in large networks is an extremely challenging task, mostly because of the large number of (potential) synapses to consider. Although we already developed a prototype, the computational resources required will became available only next year (INCITE 2018 grant). As previously discussed, we also want to improve our understanding of circuit rewiring by producing a larger experimental dataset. Our main objective will be to establish a clear link between structural and functional plasticity.

Neuromodulation and Metaplasticity are the next logical steps down the road to a unifying view of synaptic plasticity. The effects of several neuromodulators on synaptic activity are already under investigation at the Blue Brain Project (BBP). The synapse model developed in this work could easily be adapted to support these forms of plasticity. In particular, $\rho^\star$ is an ideal candidate for the extension. This parameter delimits the

basin of attraction of the potentiated and depressed state. For this reason, it directly affects the likelihood to potentiate or depress a synapse. By adding the appropriate dynamics to $\rho^\star$, it is possible to alter the default behavior of one or more synapses to implement neuron- and network-level forms of plasticity.

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Acronyms

AI Artificial Intelligence 15

AMPAR alpha-amino-3-hydroxy-5-methyl-4-isoxazole propionate receptor 20, 21, 23, 25, 26, 30–33, 38

ANN Artificial Neural Networks 15

AP action potential 19, 20, 36, 67

AZ Active Zone 20, 22

bAP backpropagating action potential 21

BBP Blue Brain Project 5, 7, 30, 31, 33, 67, 86

EPSP Excitatory Post-Synaptic Potential 23, 58, 61, 79, 86

GA Genetic Algorithm 17, 53, 55, 83

GB Graupner-Brunel 38–40, 43, 52, 85

GHK Goldman-Hodgkin-Katz 35, 37

HH Hodgkin-Huxley 34

HPC High Performance Computing 18

HVA High Voltage Activated 21, 31

LTD Long Term Depression 5, 7, 16, 17, 26, 31, 36, 40, 52, 53, 55–57, 60, 63, 83

LTP Long Term Potentiation 5, 7, 16, 17, 25, 26, 31, 33, 36, 37, 39, 40, 53, 55–57, 60, 63, 65, 83, 85

LVA Low Voltage Activated 21

MOO Multi-Objective Optimization 52

MSB Multi Synapse Bouton 21, 24

MVR Multi Vesicular Release 33, 34

NMC Neocortical Microcircuit Collaboration 34

NMDAR N-methyl-D-aspartate receptor 17, 20, 21, 23, 30–37, 39, 68, 85

NMODL NEURON model description language 66, 67

ODE Ordinary Differential Equation 32, 35, 67

PC pyramidal cell 60–63

PSD Postsynaptic Density 20, 22, 23, 56

PSP Post-Synaptic Potential 53

RRP Readily-Releasable Pool 40

SLSQP Sequential Least SQuares Programming 59

SNARE SNAP (Soluble NSF Attachment Protein) REceptor 20

STDP Spike Timing Dependent Plasticity 5, 7, 36, 38, 40, 42, 47, 49–51, 53, 56, 60, 63, 75, 83, 85

TM Tsodyks-Markram 31–33

TTPC Thick-tufted pyramidal cells 32, 40–42, 45, 47, 49–52, 58–63, 74, 75, 77, 78, 80, 81, 83

VDCC Voltage Dependent Calcium Channels 3, 20, 21, 30, 31, 34, 35, 39, 66

Glossary

me-type is a morphological and electrical type 99

pathway is the set of all connections between two given me-types 17, 18, 49, 59, 60, 63, 65, 78, 83

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